

Endocrine disruptor-induced epimutagenesis *in vitro*: Insight into molecular mechanisms

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Abstract

Endocrine disrupting chemicals (EDCs) such as bisphenol S (BPS) are xenobiotic compounds that can disrupt endocrine signaling following exposure due to steric similarities to endogenous hormones within the body. EDCs have been shown to induce disruptions in normal epigenetic programming (epimutations) that accompany dysregulation of normal gene expression patterns that appear to predispose disease states. Most interestingly, the prevalence of epimutations following exposure to many different EDCs often persists over multiple subsequent generations, even with no further exposure to the causative EDC. Many previous studies have described both the direct and prolonged effects of EDC exposure in animal models, but many questions remain about molecular mechanisms by which EDCs initially induce epimutations or contribute to the propagation of EDC-induced epimutations either within the exposed generation or to subsequent generations. Additional questions remain regarding the extent to which there may be differences in cell type-specific susceptibilities to various EDCs, and whether this susceptibility is correlative with expression of relevant hormone receptors and/or the location of relevant hormone response elements (HREs) in the genome. To address these questions, we exposed cultured mouse pluripotent (induced pluripotent stem [iPS]), somatic (Sertoli and granulosa), and germ (primordial germ cell like [PGCLCs]) cells to BPS and measured changes in DNA methylation levels at the epigenomic level and gene expression at the transcriptomic level. We found that there was indeed a difference in cell type-specific susceptibility to EDC-induced epimutagenesis and that this susceptibility correlated with differential expression of relevant hormone receptors and, in many cases, tended to generate epimutations near relevant HREs within the genome. Additionally, however, we also found that BPS can induce epimutations in a cell type that does not express relevant receptors and in genomic regions that do not contain relevant HREs, suggesting that both canonical and non-canonical signaling mechanisms can be disrupted by BPS exposure. Most interestingly, we found that when iPS cells were exposed to BPS and then induced to differentiate into PGCLCs, the prevalence of epimutations and differentially expressed genes (DEGs) initially induced in the iPSCs was largely retained in the resulting PGCLCs, however, >90% of the specific epimutations and DEGs were not conserved but were rather replaced by novel epimutations and DEGs following the iPSC to PGCLC transition. These results suggest a unique mechanism by which an EDC-induced epimutated state may be propagated transgenerationally following a single exposure to the causative EDC.

eLife assessment

The study is **solid** and reports interesting findings on the sensitivity of different cell types to EDCs *in vitro*. The data are **valuable** but the manuscript requires more experimental details. Bioinformatic analyses need to be improved and the interpretation of results and conclusions adjusted to the results.

Introduction

It has now been more than a half-century since Roy Hertz in 1958 first proposed the notion that certain chemicals, particularly those used in livestock feed at the time, could contaminate food sources and bioaccumulate in humans mimicking the activity of hormones (Gassner et al., 1958 [↗](#)). As the number of potential chemicals that could exert these effects has grown, so too has the interest in this area to the point that this is now a major topic of active research. The dangers of many endocrine disrupting chemicals (EDCs) studied to date have largely been investigated utilizing animal models. A primary example is the multi-phased Toxicant Exposures and Responses by Genomic and Epigenomic Regulators of Transcription (TaRGET) program established to determine the contribution of environmental exposures to disease pathogenesis as a function of epigenome perturbation in mouse models (Wang et al., 2018 [↗](#)). While the use of animal models has been quite informative for determining the various systemic or organ/tissue-specific disease-related effects associated with exposure to different EDCs, the molecular mechanisms underlying these phenomena remain poorly understood. Thus, questions persist regarding i) the mechanisms by which exposure of cells to EDCs results in the initial formation of epimutations defined here as any change in epigenomic programming, ii) potential cell type-specific differential susceptibility to epimutagenesis induced by different EDCs, iii) the potential involvement of relevant hormone receptors and/or genomic hormone response elements in EDC-induced disruption of the epigenome, and iv) the ability of an EDC-induced epimutated state to persist inter- or transgenerationally despite generational epigenetic reprogramming (Diaz-Castillo et al., 2019 [↗](#)).

Questions about molecular mechanisms responsible for EDC exposure-based disruption of normal epigenetic programming have been difficult to interpret in whole animal models due, in part, to the inherent complexity of the intact animal including potential interactions among multiple different organs, tissues, and cell types at the paracrine, metabolic and/or systemic levels. To circumvent this challenge, we opted to expose homogeneous populations of specific cell types in culture to doses of the EDC, BPS, that are below the maximum safe limit previously set by the EPA for the similar estrogen mimetic, BPA. In this way we hoped to learn more about the manner in which direct exposure to a specific EDC induces epimutations and differential gene expression in different individual cell types including somatic cells, pluripotent cells, and germ cells. This approach was designed to distinguish the direct effects of an EDC exposure *in vitro* from indirect effects that may accrue as the result of an EDC exposure on one cell type inducing a secondary effect on a neighboring or related cell type *in vivo*.

Previous *in vitro* studies have demonstrated direct susceptibility of cultured cell types to EDC-induced epimutagenesis, however, those studies were focused on immortalized cancer cell lines that do not model key cell types involved in normal initial incursion or subsequent intragenerational propagation or inter- or transgenerational transmission of environmentally-induced epimutations *in vivo* (Deb et al., 2016 [↗](#); Goodman et al., 2014 [↗](#); Huang et al., 2019 [↗](#); Karaman and Ozden, 2019 [↗](#); Senyildiz et al., 2017 [↗](#)). We chose to examine direct exposure of three key types of cells relevant to environmental exposures and subsequent propagation and

transmission of induced epimutations *in vivo* – somatic cell types known to be responsive to endocrine signaling (Sertoli cells and granulosa cells), pluripotent cells (iPSCs) mimicking the preimplantation embryo, and germline cells (PGCLCs).

While there have been reports that have clearly indicated the ability of EDCs to disrupt classical endocrine signaling (Henley and Korach, 2010; Kelce et al., 1995 [↗](#); Swedenborg et al., 2009 [↗](#); vom Saal and Hughes, 2005 [↗](#); You et al., 1998 [↗](#)), there exist other reports indicating EDCs can also disrupt non-classical endocrine signaling via nuclear receptors (Ozgyin et al., 2015 [↗](#); Song et al., 2011 [↗](#)), G protein-coupled receptors (Thomas and Dong, 2006 [↗](#)), and calcium channel signaling (Brenker et al., 2018 [↗](#)). This has further complicated our understanding of the mechanism(s) by which EDC exposure initially induces epimutations, and has particularly confounded insight into the susceptibility of germ cells to EDC-induced epimutagenesis given that germ cells are reported to lack expression of classical endocrine receptors (Meccariello et al., 2014 [↗](#)). This is relevant to the effort to understand mechanisms underlying transgenerational epigenetic inheritance of EDC-induced epimutations (Anway et al., 2006 [↗](#); Guerrero-Bosagna et al., 2012 [↗](#); Nilsson et al., 2008 [↗](#)) which clearly implicates transmission via the germ line.

The concept of germline transmission of EDC-induced epimutations is further confounded by the known epigenetic reprogramming events that occur in the preimplantation embryo and developing fetal and neonatal germ line. Thus, regardless of how EDC-induced epimutations become initially manifest in the germ line, it is not clear how they subsequently persist and are transmitted inter- or transgenerationally given the large portion of the epigenome that undergoes genome-wide erasure and resetting of epigenetic programming during each generation (Cantone and Fisher, 2013 [↗](#); Lee et al., 2014 [↗](#); Santos et al., 2002 [↗](#); Sanz et al., 2010 [↗](#)). *In vitro* cell culture systems afford the opportunity to focus on individual cell types independently, including somatic cells which are likely initially exposed to most environmental disruptive influences, pluripotent cells which represent the preimplantation embryo in which the first major reprogramming event occurs during normal development, and early germline cells where the second major reprogramming event is initiated in primordial germ cells (PGCs). Importantly, in addition to the potential to study each of these cell types independently, it is possible to induce transitions in cell fate *in vitro* that model those that occur during normal development, thereby recapitulating normal reprogramming events in a way that can facilitate high resolution studies of the fate of EDC-induced epimutations once they have been induced in any of these cell types.

Here we describe our study of the relative susceptibility to induction of epimutations by direct exposure of four different cell types maintained in culture – somatic Sertoli and granulosa cells, pluripotent iPSCs and germline PGCLCs – to BPS, followed by our analysis of the fate of epimutations initially induced in mouse iPSCs that are then induced to transition into PGCLCs. We found that there are cell type-specific differences in susceptibility to epimutagenesis and associated dysregulation of gene expression patterns following exposure of these different cell types to a similar dose of BPS below the maximum safe exposure limit established for BPA by the EPA (Ribeiro et al., 2019 [↗](#)). We further found that these differences correlate at least partially with differential expression of relevant hormone receptors, although germ cells, which do not express relevant receptors, also incurred epimutations following direct exposure to BPS, albeit less extensively than somatic or pluripotent cell types. In all cell types tested, exposure to BPS disrupted epigenetic programming in potentially important regulatory regions including enhancers and promoters. Beyond this, however, we found that the distribution of many of the epimutations induced by exposure to BPS, which is an estrogen mimetic, correlated with the genomic locations of estrogen response elements (EREs) in cell types expressing estrogen receptors, but much less so in germ cells that do not express estrogen receptors. To address the origin of these BPS-induced epimutations in PGCLCs, we identified G-protein coupled receptor targets that potentially could be involved in non-canonical estrogen signaling and susceptibility to BPS exposure. Most interestingly, we found that when iPSCs exposed to BPS were then induced to undergo a major transition in cell fate and related epigenetic reprogramming to form PGCLCs, a

similar prevalence of epimutations and DEGs was retained throughout this transition, but very few (<10%) specific epimutations or DEGs were conserved during this process. This suggests that the initial EDC exposure induces a disruption of the chromatin landscape that subsequent epigenetic reprogramming is unable to fully restore. Thus, reprogramming during a major cell fate transition involving cells exposed to BPS appears to correct many of the initially induced epimutations, but also appears to induce many *de novo* epimutations, and this imbalance may persist across multiple generations. This represents a novel mechanism that may underlie continued transgenerational epigenetic inheritance of a prevalence of EDC-induced epimutations despite ongoing generational reprogramming during succeeding generations.

Results

Dose dependent epimutagenesis response to BPS exposure in iPSCs

To initially assess the extent to which epimutations and dysregulated gene expression can be induced in cells maintained in culture, we exposed mouse iPSCs to three doses of BPS (1, 50, and 100 μ M) and measured the impact on the epigenome and transcriptome using the Illumina Infinium Mouse Methylation BeadChip Array and RNA-seq, respectively (**Figure 1** [↗](#)). All three doses of BPS induced individual differentially methylated CpGs (DMCs) as well as differentially methylated regions (DMRs) (**Figure 1a** [↗](#)), and differentially expressed genes (DEGs) (**Figure 1b** [↗](#)) when compared to control mouse iPSCs treated with vehicle (EtOH) only. The extent of this exposure-specific epimutagenesis was directly correlated with the dose of BPS used. These results provided an initial proof of concept that exposure of cells maintained *in vitro* to an EDC such as BPS can disrupt the epigenome and transcriptome in a dose-dependent manner. Interestingly, exposure to 1 μ M BPS induced very few DMRs, but did induce widespread DMCs and DEGs. Because 1 μ M BPS is below the FDA's suggested safe environmental level established for BPA (Ribeiro et al., 2019 [↗](#)), and was sufficient to induce DMCs and dysregulated gene expression on all chromosomes, we utilized this dose and focused solely on DMCs when assessing epimutations in all subsequent experiments.

Cell type-specific susceptibility to induction of epimutations following BPS exposure

We next sought to determine if different key cell types – somatic, pluripotent and germ – are differentially susceptible to induction of epimutations in response to direct exposure of each to a similar dose of BPS. Thus, we exposed pluripotent (iPSCs), somatic (Sertoli and granulosa cells), and germ (PGCLCs) cell types to 1 μ M BPS and measured changes in DNA methylation patterns. We identified exposure-specific DMCs in each exposed cell type relative to its corresponding control (exposed to carrier only) (**Figure 2a** [↗](#)). We observed overall differences among the different cell types in total numbers of DMCs, with iPSCs showing the highest number of DMCs, followed by Sertoli cells and granulosa cells, and then PGCLCs, respectively (**Table 1** [↗](#)).

Interestingly, among the exposure-specific DMCs identified in each cell type, we observed predominantly hypomethylated DMCs in the somatic and pluripotent cell types, but predominantly hypermethylated DMCs in PGCLCs (**Table 1** [↗](#)). These findings confirm that there are both quantitative differences among cell types in overall susceptibility to epimutagenesis, and qualitative cell type-specific differences in the prevalence of hypo-versus hypermethylated DMCs following exposure to BPS. The latter likely reflects the fact that the epigenome in PGCs is naturally more hypomethylated than that in pluripotent or somatic cell types (Hajkova, 2011 [↗](#)), enhancing the likelihood that most changes in DNA methylation induced in PGCLCs will necessarily involve hypermethylation.

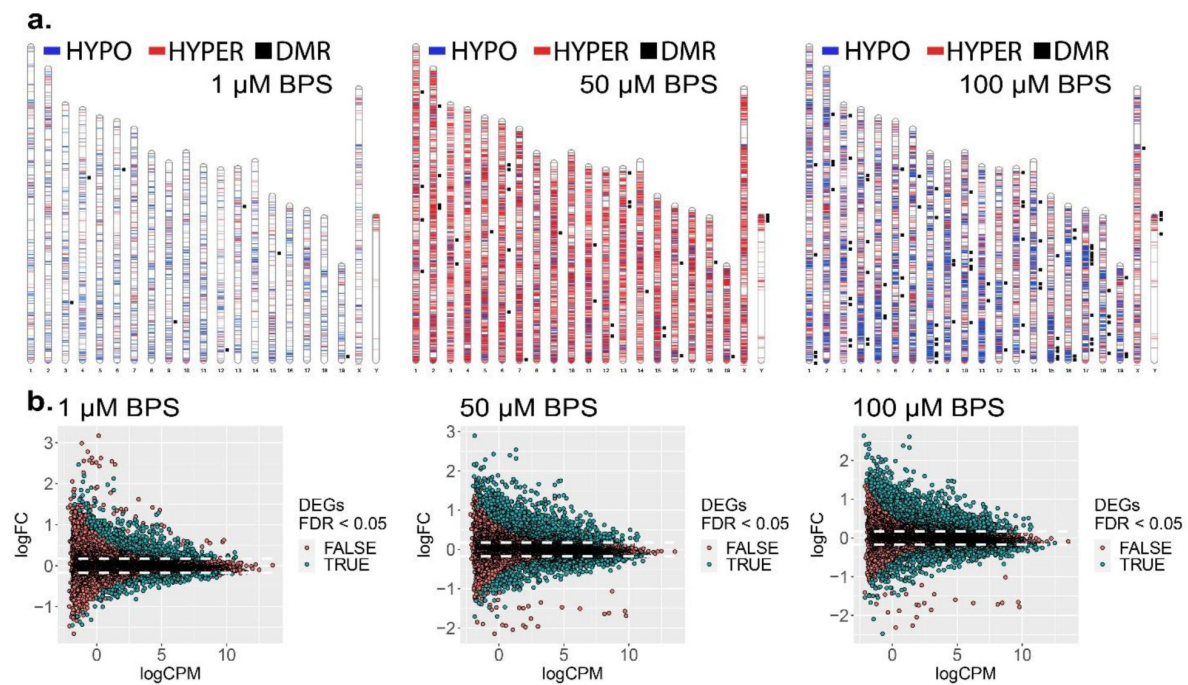


Figure 1.

Dose dependent impact of epimutagenesis measured in iPSCs exposed to 1, 50, and 100 μ M BPS.

a) Ideogram plots displaying chromosomal distribution of genome-wide changes in DNA methylation caused by BPS exposure. **b)** Mean difference (MD) plots of changes in gene expression following exposure to increasing doses of BPS. Exposure to increasing doses of BPS induced corresponding increases in the numbers of DMCs, DMRs, and DEGs. Blue horizontal lines = hypomethylated DMCs, red horizontal lines = hypermethylated DMCs, black squares = DMRs.

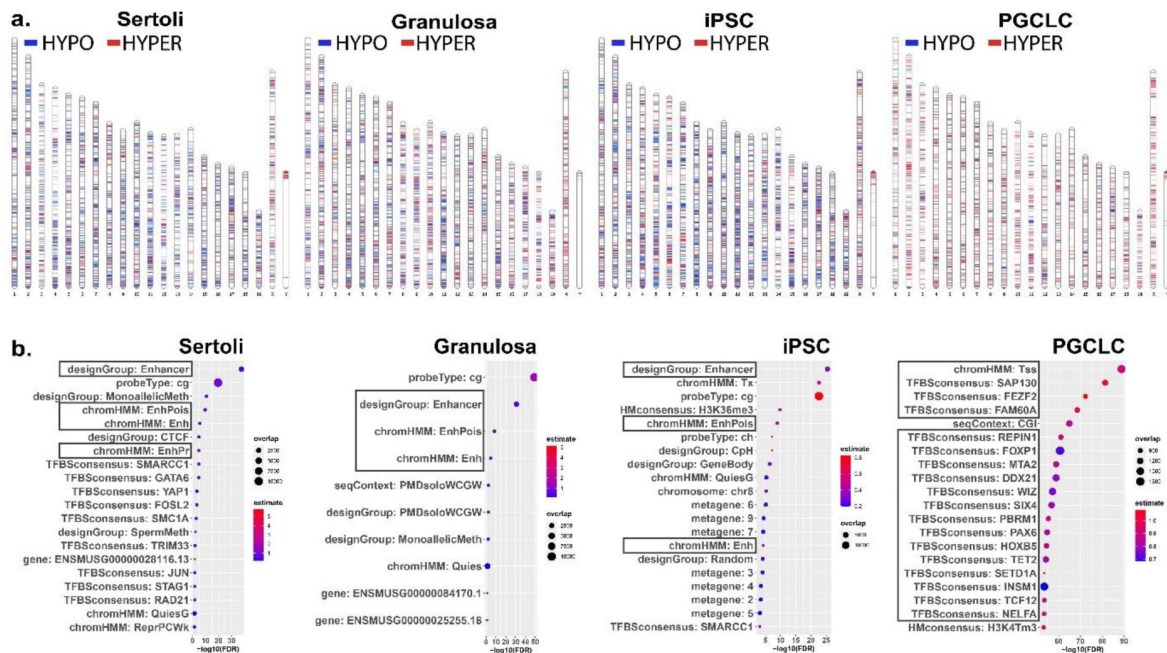


Figure 2.

Chromosomal distributions and annotations of BPS-induced epimutations in pluripotent, somatic, and germ cell types.

a) Ideograms illustrating chromosomal locations of DMCs induced by exposure of each cell type to 1 μ M BPS. Blue horizontal lines = hypomethylated DMCs, red horizontal lines = hypermethylated DMCs. **b)** Enrichment plots indicating feature annotations in genomic regions displaying prevalent BPS-induced epimutations in each cell type.

DMCs	Sertoli	Granulosa	iPSCs	PGCLCs
Hypomethylated ¹	7,385	6,444	9,651	2,315
Hypermethylated ²	3,022	4,143	4,308	4,785
Total	10,407	10,587	13,959	7,100

¹A CpG site that was predominantly methylated in the control samples but unmethylated in the exposed samples.

²A CpG site that was predominantly unmethylated in the control samples but methylated in the exposed samples.

Table 1.

Treatment-specific differentially methylated sites (DMCs) (treated vs. control)

Annotation of BPS-induced epimutations

We next analyzed annotations associated with genomic sites of BPS exposure-specific DMCs in each cell type. We mined annotations associated with genomic regions included on the Infinium array and found that a substantial group of exposure-specific DMCs was associated with enhancer regions in the two somatic (Sertoli and granulosa) and one pluripotent (iPS) cell types exposed to BPS, while in PGCLCs BPS-induced DMCs were more prevalent at promoter regions containing transcription start sites (TSSs) and transcription factor binding sites (**Figure 2b** [↗](#)), indicative of yet another qualitative cell type-specific difference in induction of epimutations by the same dose of BPS.

Relationship between susceptibility to BPS-induced epimutagenesis and expression of relevant hormone receptors

As EDCs are thought to induce epimutations via disruption of classical hormonal signaling (Henley and Korach, 2010; Kelce et al., 1995 [↗](#); Swedenborg et al., 2009 [↗](#); vom Saal and Hughes, 2005 [↗](#); You et al., 1998 [↗](#)), we next sought to determine if the differential extent of BPS induction of epimutations we observed in different cell types was associated with differential expression of relevant hormone receptors in each. We performed immunocytochemistry (ICC) to detect the presence of the relevant estrogen receptors – ER α and ER β , while co-staining for cell type-specific markers (WT1 [Sertoli cell marker], FSHR [granulosa cell marker], FUT4 [iPSC marker], and NANOG [PGCLC or endogenous PGC marker]) to confirm the identity of the four cultured cell types examined in this study (Sertoli cells, granulosa cells, iPSCs and PGCLCs), as well as endogenous mouse PGCs (**Figure 3a** [↗](#)). Both somatic cell types (Sertoli and granulosa) showed positive immunolabeling for both ER α and ER β , whereas the pluripotent cells showed positive immunolabeling for ER β , but negative immunolabeling for ER α , and the PGCLCs and endogenous PGCs showed negative immunolabeling for both ER α and ER β . Importantly, the latter result is consistent with previous reports of lack of expression of either ER α or ER β at the protein level in endogenous PGCs (Meccariello et al., 2014 [↗](#)).

These results demonstrate cell type-specific differences in expression of hormone receptors (ER α and ER β) which are potentially relevant to the disruptive action of the estrogen mimetic, BPS. We noted that expression of at least one potentially relevant hormone receptor (ER β) in either somatic or pluripotent cells correlated with a higher incidence of DMCs induced by exposure of somatic or pluripotent cell types to 1 μ M BPS relative to the incidence of DMCs induced by exposure of PGCLCs, which do not express either estrogen receptor, to the same dose of BPS (**Table 1** [↗](#)). Nevertheless, we did still observe induction of DMCs when PGCLCs were exposed to BPS, demonstrating that expression of relevant canonical hormone receptors is not an absolute requirement for induction of epimutations in response to an EDC. As noted above, we also observed differences in annotations of the genomic regions in which BPS induced epimutations in germ cells relative to the other cell types tested, with ER α - and/or ER β -expressing somatic and pluripotent cell types showing a higher incidence of epimutations at apparent enhancer regions while ER α /ER β -negative germ cells showed a higher epimutation incidence at apparent promoter regions (900 bp upstream and 500 bp downstream from TSS) (**Figure 2b** [↗](#)). Taken together, these results reveal cell type-specific differences in susceptibility to induction of epimutations by exposure to the EDC, BPS.

Proximity of BPS-induced epimutations to genomic EREs

The relatively higher incidence of BPS-induced epimutations in cell types expressing one or both estrogen receptor(s) suggests BPS-induced epimutagenesis may be manifest, at least in part, through canonical endocrine signaling pathways. If this is the case, we might expect to see elevated induction of epimutations in genomic regions enriched for relevant HREs which, for interactions with the estrogen mimetic, BPS, would be EREs. Previous studies have defined a full

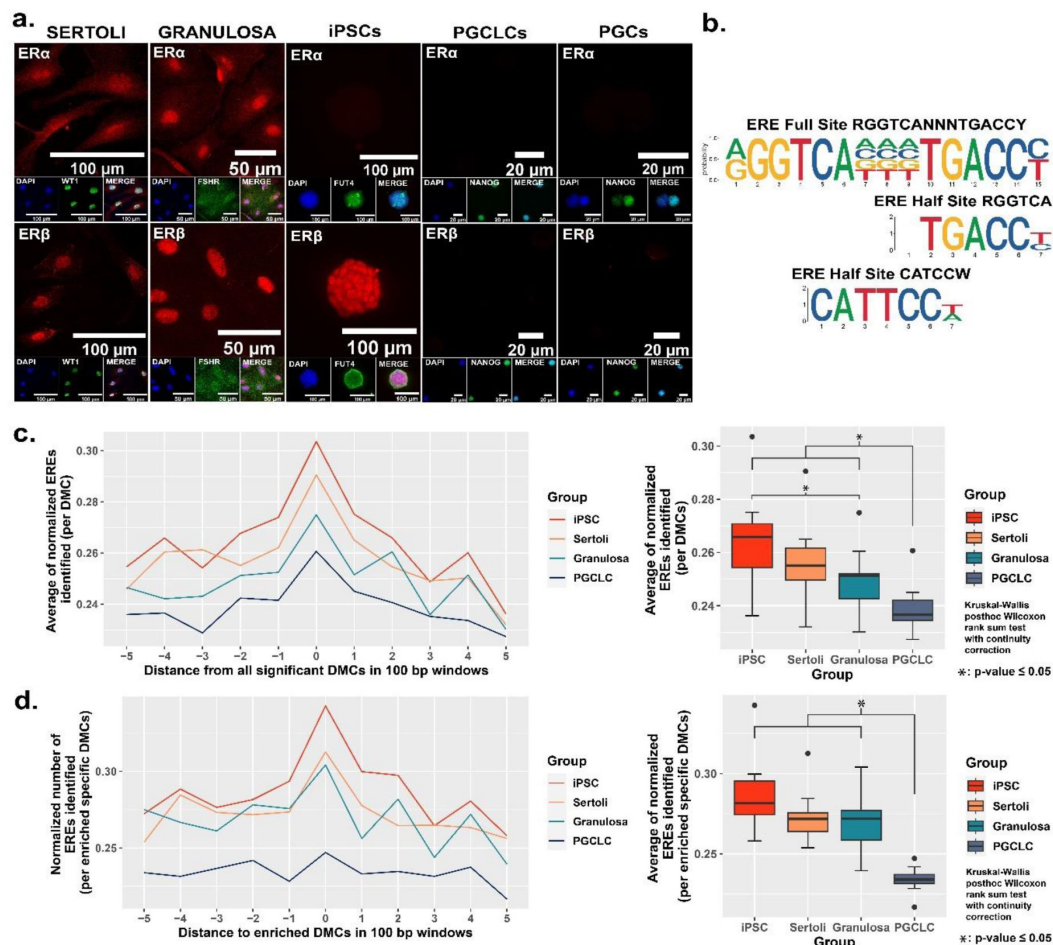


Figure 3.

Correlation between cell type-specific expression of estrogen receptors and density of genomic EREs associated with BPS-induced epimutations.

a) Assessment of expression of ERα and ERβ on cell types co-stained for known cell type-specific markers. Somatic cell types express both receptors, pluripotent cells express ERβ but not ERα, and germ cells do not express either estrogen receptor. **b)** Motif plots displaying the full ERE consensus sequence and the more biologically relevant ERE half-site motifs found to be enriched from ERα ChIP-seq. **c,d)** Normalized density plots and box plots displaying the frequency of ERE half-sites identified c) within 500 bp of all BPS-induced DMCs genome-wide, or d) within 500 bp of the most enriched categories of BPS-induced DMCs in each cell type (= enhancer regions for somatic and pluripotent cell types and promoter regions in the germ cell type).

ERE consensus sequence (Bourdeau et al., 2004 [↗](#)), but other reports have indicated that estrogen receptors can often bind to ERE half-sites (Mason et al., 2010 [↗](#)) (**Figure 3b** [↗](#)). Indeed, when we mined publicly available ERα ChIP-seq peaks from the UCSC genome browser database (Dunham et al., 2012 [↗](#); Myers et al., 2011 [↗](#); Sloan et al., 2016 [↗](#); Wang et al., 2013 [↗](#), 2012 [↗](#)) and performed motif enrichment, we identified two distinct ERE half-sites within regions enriched for DMCs rather than one full sized ERE consensus sequence, with the half-sites appearing to be more biologically relevant to interaction with estrogen or its mimetics (**Figure 3b** [↗](#)). We then plotted the frequency of ERE half-sites in genomic regions within 500 bp of BPS-induced epimutations (**Figure 3c,d** [↗](#)). We found an increase in the frequency of ERE half-sites identified within 500 bp of BPS-induced DMCs genome-wide in all four cell types investigated, although the frequency was notably lower in germ cells (**Figure 3c** [↗](#)). Interestingly, this higher frequency of ERE half-sites within 500 bp of BPS-induced DMCs was conserved in three of the four cell types – somatic (Sertoli and granulosa) and pluripotent (iPSCs) when we focused solely on apparent enhancer regions, but not in germ cells (PGCLCs) where the majority of DMCs occurred at apparent promoter regions (**Figure 3d** [↗](#)). We also mined the UCSC mouse genome sequence for genome-wide prevalence of ERE half-sites and found that while these sites occur in both promoter and enhancer regions, they are nearly four-fold more frequent in enhancers (Supplementary file 1b). Thus, it is perhaps not surprising that BPS-induced epimutations were more prevalent in enhancer regions in somatic and pluripotent cell types that express one or both ERs, than in germ cells that do not express either ER.

Taken together, these results suggest a relationship between i) expression of either ERβ alone or ERα and ERβ together, ii) elevated susceptibility to BPS-induced epimutagenesis, and iii) the occurrence of BPS-induced DMCs in genomic regions – particularly enhancers – containing EREs. These observations are consistent with the notion that one mechanism contributing to EDC-based induction of epimutations involves disruption of canonical endocrine signaling pathways. However, the fact that exposure to BPS also induced epimutations in germ cells, even in the absence of expression of estrogen-related receptors, and generated many DMCs in regions not inclusive of EREs, suggests that disruption of canonical endocrine signaling pathways is not the only mechanism by which exposure to an EDC can induce epimutations.

Cell type-specific features of BPS-induced epimutations

To further interrogate cell type-specific differences in the genesis of epimutations following exposure to BPS, we compared DNA methylation patterns detected in the control (vehicle only) samples to identify naturally occurring, cell type-specific DMCs inherently associated with each different cell fate. Interestingly, of the 297,415 distinct CpGs interrogated by the Illumina Infinium Mouse Methylation BeadChip Array used for this analysis, >240,000 showed some degree of inherent differential methylation among the four cell types tested, suggesting they are tied to cell-fate specific differential DNA methylation (**Figure 4a** [↗](#)). We next determined the extent to which DMCs induced specifically by exposure of each cell type to BPS occurred at CpG dinucleotides that were among these naturally occurring cell type-specific DMCs. We found that a large majority of BPS exposure-specific epimutations detected in each cell type occurred at CpGs that also show inherent, cell type-specific variation in DNA methylation (>95% of BPS-induced epimutations in somatic and pluripotent cell types and ~89% in germ cells) (**Figure 4a** [↗](#)). As with the overall pattern of BPS-induced epimutations shown in **Figure 2** [↗](#), BPS-induced epimutations at CpGs showing inherent cell type-specific variation were enriched in apparent enhancer regions that occurred near EREs in somatic and pluripotent cell types, whereas those in germ cells were found predominantly in promoter regions containing significantly fewer EREs (**Figure 4b,d** [↗](#)). However, the low percentage of BPS-induced epimutations that occurred at CpGs that did not show inherent cell type-specific variation in DNA methylation patterns were found to be enriched in promoter regions lacking nearby EREs in all four cell types (**Figure 4c,d** [↗](#)). Thus, this latter group of BPS-induced epimutations appears to represent a small core group that arises following exposure to

BPS via a mechanism that does not rely upon disruption of canonical endocrine signaling, and so is common to all cell types, regardless of expression of relevant endocrine receptors or nearby residence of relevant HREs within the genome.

Impact of BPS exposure on gene expression

We next sought to determine if the cell type-specific differential susceptibility to BPS-induced epimutagenesis translated to a similar extent of dysregulation of gene expression in each cell type.

Somewhat surprisingly, RNA-seq analysis of gene expression patterns in each exposed cell type relative to its corresponding control (same cell type exposed to carrier only) revealed the greatest number of dysregulated genes in PGCLCs, despite PGCLCs being the cell type that showed the lowest number of epimutations following exposure to BPS (Tables 1,2). iPSCs showed the second highest level of dysregulated genes, while somatic Sertoli and granulosa cells showed very low levels of dysregulated gene expression (**Table 2** [↗](#)). Indeed, numbers of dysregulated genes were three orders of magnitude lower in differentiated somatic cells than in either pluripotent cells or germ cells (**Table 2** [↗](#)).

Because BPS-exposed PGCLCs showed the highest level of dysregulated genes, as well as the highest enrichment of DMCs occurring primarily at promoters (**Figure 2b** [↗](#)), we next assessed the general proximity between DMCs and promoter regions in each cell type to determine if DMCs in promoter regions may be more likely to predispose dysregulated gene expression than DMCs elsewhere in the genome (**Figure 5a** [↗](#)). Indeed, we found that in all four cell types, the smaller the median distance between BPS-induced DMCs and neighboring promoter regions, the larger the number of BPS-induced DEGs (**Figure 5b** [↗](#)). Thus, it appears that while PGCLCs showed the fewest BPS-induced DMCs among the four cell types exposed to BPS, this exposure induced a higher proportion of epimutations in regions within or adjacent to promoters in this cell type. This, and the relatively high extent of decondensed chromatin genome-wide in fetal germ cells, appear to have contributed to the higher incidence of dysregulated gene expression in BPS-exposed PGCLCs than that observed in the other three BPS-exposed cell types, even though the overall numbers of BPS-induced epimutations were greater in the other cell types.

Potential non-canonical signaling pathways disrupted by BPS exposure

To identify potential mechanisms by which BPS exposure may induce epimutations via disruption of non-canonical endocrine signaling pathways, we mined our RNA-seq data to identify genes dysregulated independently of expression of relevant hormone receptors or presence of nearby EREs.

We identified a set of genes in all four cell types tested that displayed promoters enriched for apparent endocrine-signaling independent DMCs (Supplementary file 1a). Gene ontology (GO) analysis indicated that several of the 1,957 genes we identified were associated with ubiquitin-like protease pathways, including 124 of 800 genes that regulate protein degradation and 5 of 9 genes involving 1-phosphatidylinositol-3 kinase activities, linked to the PI3K/AKT signaling pathway (**Figure 6a** [↗](#)). Previous reports have established a link between ubiquitin-like protease pathways and classical ER signaling (Beamish and Frick, 2022 [↗](#); Kabir et al., 2015 [↗](#)), suggesting that genes with promoters lacking EREs may be regulated by factors encoded by genes with promoters containing EREs, and so could be indirectly activated or repressed by disruption of classical ER signaling. Further, involvement of PI3K/AKT pathway signaling has previously been linked to estrogen signaling through G-protein coupled receptors such as GPER1, which was implicated in non-canonical estrogen signaling in ER KO mouse cell lines (Filardo et al., 2002 [↗](#); Molina et al., 2017 [↗](#)). To determine if our data support this suggestion, we performed GO analysis on the 1,890

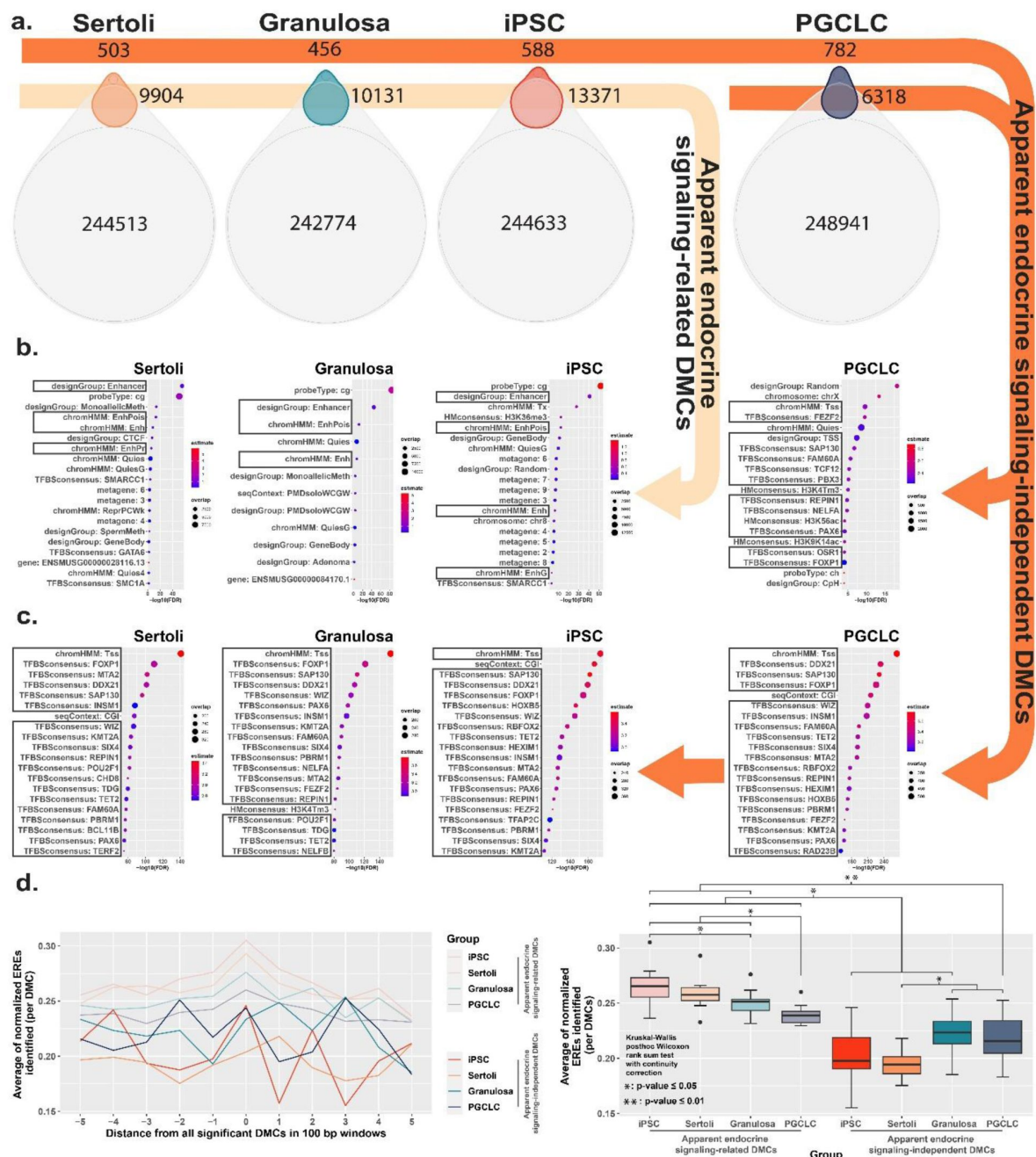


Figure 4.

Direct comparison of BPS exposure-specific and cell type-specific features between cell types.

a) Assessment Venn diagrams indicating DMCs that are due either to BPS exposure (top, smaller ovals) or inherent cell type-specific differences (bottom, larger ovals). Numbers of apparent endocrine signaling-related DMCs are shown in the light orange arrow, and apparent endocrine signaling-independent DMCs are shown in the dark orange arrows. Enrichment plots indicating feature annotations in genomic regions displaying

b) apparent endocrine signaling-related DMCs occurring predominantly in enhancer regions in somatic Sertoli and granulosa cell types or pluripotent cells expressing one or more estrogen receptors, or

c) a smaller set of apparent endocrine signaling-independent DMCs occurring predominantly in promoter regions in all four cell types regardless of +/- expression of relevant endocrine receptors.

d) Normalized density plots and box plots displaying the frequency of ERE half-sites identified within 500 bp of apparent endocrine signaling-related DMCs occurring predominantly in enhancer regions and apparent endocrine signaling-independent DMCs occurring predominantly in promoters.

DEGs	Sertoli	Granulosa	iPSCs	PGCLCs
Down regulated	3	0	343	844
Up regulated	32	2	694	1,046
Total	35	2	1,037	1,890

*Genes showing significant differential expression following exposure of each cell type to 1μM BPS relative to matched control cell types exposed to carrier only.

Table 2.

Exposure-specific differentially expressed genes*

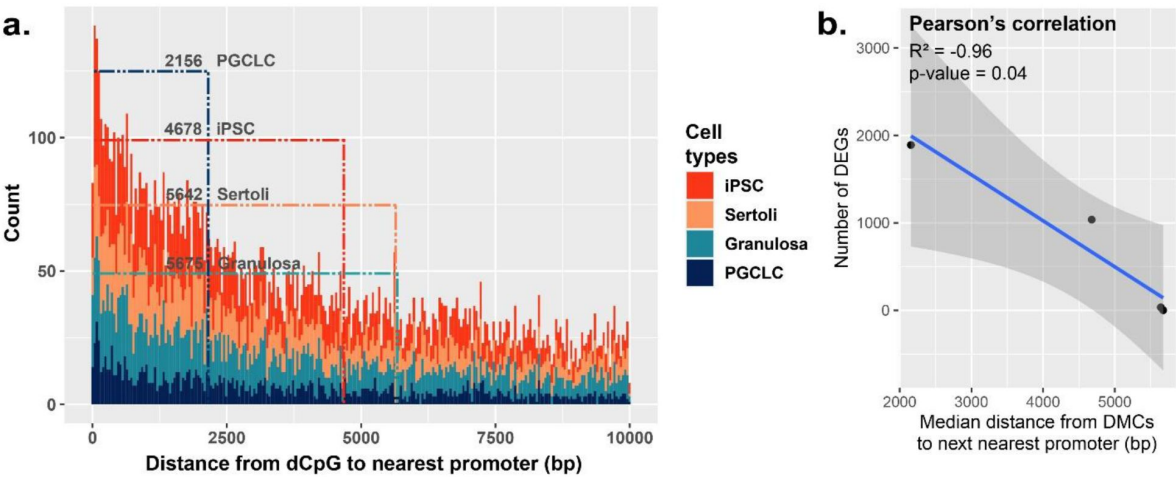


Figure 5.

Correlation between the proximity of DMCs to promoters and dysregulation of gene expression.

a) Proximity plot displaying distances from exposure-specific DMCs to nearest promoter regions. Dotted lines indicate median points of the data for each cell type. **b)** Correlation plot displaying a negative relationship between the distance from DMCs to nearest promoters and resulting dysregulation of gene expression within each cell type.

DEGs identified in PGCLCs in which epimutations appeared to be induced independent of classical ER signaling. We detected no apparent involvement of pathways involving ubiquitin-like proteases, but we did detect differential expression of 4 of 7 genes associated with a pathway involving phosphatidylinositol-4 phosphatase signaling which intersects with the PI3K/AKT signaling pathway (**Figure 6b**). Finally, while we did not detect differential expression of *Gper1* transcripts, we did observe differential expression of genes encoding other less well studied G-protein coupled receptors, including *Gprc5a*, *Gprc5b*, *Gpr89*, *Gpr107*, and *Gpr161* in all four cell types exposed to BPS. These receptors have all been previously shown to bind either BPA or 17 β -estradiol in rat models (<https://rgd.mcg.edu/rgdweb/report/gene/main.html?id=1305882>, <https://rgd.mcg.edu/rgdweb/report/gene/main.html?id=1563245>, <https://rgd.mcg.edu/rgdweb/report/gene/main.html?id=1304837>, <https://rgd.mcg.edu/rgdweb/report/gene/main.html?id=1310804>, <https://rgd.mcg.edu/rgdweb/report/gene/main.html?id=1309510>) and could be potential targets of non-canonical BPS-induced estrogen signaling (**Figure 6c**).

Propagation of BPS-induced epimutations during transitions in cell fate

Transitions between pluripotent and germ cell fates, or vice versa, are accompanied by large-scale epigenetic reprogramming *in vivo* (Cantone and Fisher, 2013; Lee et al., 2014; Santos et al., 2002; Sanz et al., 2010), and these are recapitulated during similar transitions induced *in vitro* (Ishikura et al., 2016). To determine the extent to which BPS-induced epimutations persist during a pluripotent to germline transition *in vitro*, we first exposed iPSCs to 1 μ M BPS and then differentiated the exposed iPSCs first into epiblast-like cells (EpiLCs) and then into PGCLCs recapitulating the early germline epigenetic reprogramming event that normally occurs *in vivo* (Kurimoto and Saitou, 2018; Ohta et al., 2017) (**Figure 7a**). We then compared the numbers of exposure-specific DMCs and DEGs detected in the PGCLCs derived from the BPS-exposed iPSCs with those in the directly exposed iPSCs and found lower, but still substantial numbers of both (28,168 vs 38,105 DMCs, and 1,437 vs 1,637 DEGs in the derived PGCLCs and directly exposed iPSCs, respectively) (**Figure 7b,c**). However, when we compared the specific DMCs and DEGs that were detected in the BPS-exposed iPSCs with those that were detected in the subsequently derived PGCLCs, we found that >90% of each were not conserved during the pluripotent to germline transition in cell fate. Specifically, only 3.7% of the DMCs and 8.4% of the DEGs detected in the BPS-exposed iPSCs were also detected in the PGCLCs derived from the exposed iPSCs. Thus, of the 38,105 DMCs and 1,637 DEGs induced by exposure of iPSCs to 1 μ M BPS, 36,688 and 1,499, respectively, did not persist during differentiation of the iPSCs to form PGCLCs and so were apparently corrected by germline reprogramming. Simultaneously however, 26,752 novel DMCs and 1,299 novel DEGs appeared in the PGCLCs that were not present in the BPS-exposed iPSCs, so were apparently generated *de novo* during the germline reprogramming process. These results are consistent with the notion that germline epigenetic reprogramming corrected many of the epimutations that were present in the BPS-exposed pluripotent cells from which they were derived, but that exposure of cells to EDCs may disrupt the underlying chromatin landscape in a way that then interferes with subsequent reprogramming such that in addition to correcting many previously induced epimutations, the germline reprogramming process, acting on a disrupted chromatin landscape, also generates many novel epimutations *de novo* during the pluripotent to germline transition.

Discussion

While nearly 20 years of research on the effects of exposure of live animals to various EDCs and other environmental disruptive influences has clearly established the potential to perturb the epigenome in ways that can dysregulate normal gene expression patterns and predispose the development of disease states, the molecular mechanisms underlying these phenomena have

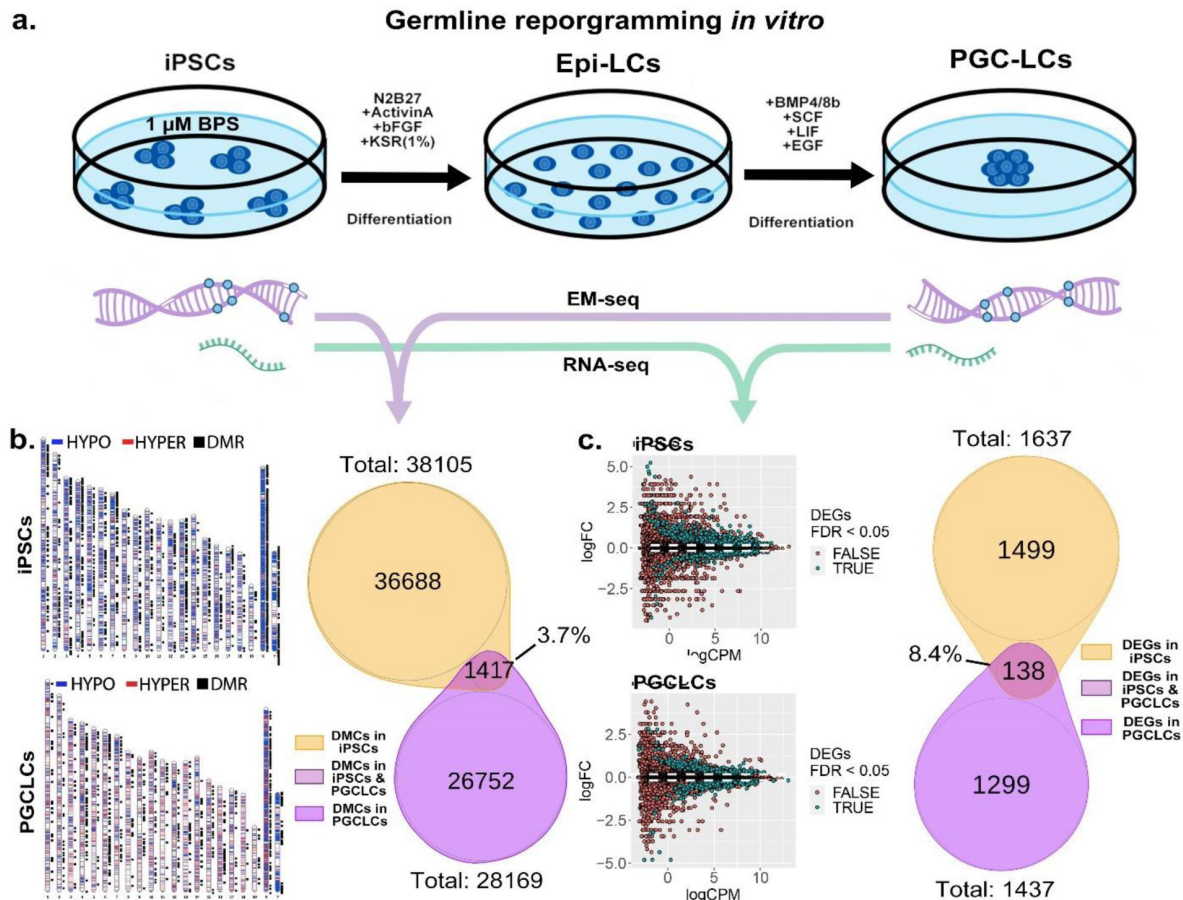


Figure 7.

Persistence of BPS-induced epimutations through recapitulation of early germline reprogramming *in vitro*.

a) Schematic illustrating derivation of PGCLCs from iPSCs *in vitro*. iPSCs are first induced to form EpiLCs which are then induced to form PGCLCs. iPSCs were exposed to either ethanol + 1 μ M BPS or ethanol (carrier) only, then induced to undergo transitions to form EpiLCs and then PGCLCs. **b)** DNA samples from BPS-exposed or control iPSCs as well as subsequently derived PGCLCs were assessed for DNA methylation epimutations by EM-seq. BPS-treated iPSCs showed 38,105 DMCs and subsequently derived PGCLCs showed 28,169 DMCs. Of those, only 1,417 (3.7%) of the DMCs were conserved from the BPS-exposed iPSCs to the subsequently derived PGCLCs. **c)** RNA samples from BPS-exposed or control iPSCs and subsequently derived PGCLCs were assessed for global gene expression patterns by RNA-seq. BPS-treated iPSCs showed 1,637 exposure-specific DEGs and subsequently derived PGCLCs showed 1,437 exposure-specific DEGs. Of those, only 138 (8.4%) were conserved from the BPS-exposed iPSCs to the subsequently derived PGCLCs.

remained largely undefined. Thus, the manner in which environmental exposures introduce biochemical changes in epigenetic programming, how such disruptions can be propagated within a tissue or throughout the soma of an exposed individual or enter that individual's germ line, or how these disruptions predispose disease states in that individual, or how a resulting prevalence of epimutations can be transmitted to multiple subsequent generations despite generational epigenetic reprogramming, are all mechanistic questions that have remained unanswered despite more than 5,000 publications on this topic since the early 2000s. The vast majority of those publications have described experiments in animal models – typically rodents. While informative, such live animal studies require months to years to complete, are cost- and animal-intensive, and have not yielded insight into the molecular mechanisms underlying the deleterious effects of environmentally-induced epimutations.

In vitro model systems have proven to be very useful tools for deciphering molecular mechanisms related to normal development and homeostasis, or disruptions of those processes predisposing onset of disease. Examples include cell culture systems used to study developmental processes such as X-chromosome inactivation in mammals (Almeida et al., 2017 [↗](#); Patrat et al., 2009 [↗](#)) or early embryogenesis (Bao et al., 2022 [↗](#); Lau et al., 2022 [↗](#)), or to elucidate the cellular and molecular etiology of many different diseases based on the now popular “disease-in-a-dish” approach (Davaapil et al., 2020 [↗](#); Song et al., 2023 [↗](#)). We assembled an *in vitro* system in which we combined the three cell types normally involved in i) the initial exposure to disruptive environmental effects (somatic cells), ii) the first phase of generational epigenetic reprogramming in the preimplantation embryo (pluripotent cells), and iii) the second phase of generational reprogramming in the developing germ line (germ cells). This system allowed us to gain insights into multiple mechanistic aspects of environmentally-induced epimutagenesis including the initial induction of epimutations by exposure of cells to the EDC, BPS, cell type-specific differences in susceptibility to BPS-induced epimutagenesis, factors that appear to contribute to that differential susceptibility, and the mechanism by which an epimutated state is transmitted across multiple generations despite generational reprogramming.

Here we have shown that exposure of homogeneous populations of cells maintained in culture to an EDC such as BPS can induce epimutagenesis and dysregulation of gene expression, and that this occurs in a dose-dependent manner. This afforded us the opportunity to test the relative effects of direct exposure of different specific cell types to a similar dose of BPS with no confounding effects imposed by interactions with other cell types or systemic effects of the sort that normally occur in the *in vivo* context. We found that somatic, pluripotent and germ cell types do indeed show differences in direct susceptibility to induction of epimutations following exposure to BPS, both in terms of the quantity of epimutations induced and qualities of the resulting epimutations such as a predominance of hyper-versus hypomethylation, and the genomic locations where such epimutations tend to occur. Our results suggest that these differences reflect distinctions in the normal status of each cell type at the time of exposure. Thus, somatic, pluripotent and germ cell types differ in the normal status of genome-wide patterns of accessible versus inaccessible chromatin, global DNA methylation, and expression of relevant endocrine receptors. Interestingly, our data suggest that all of these parameters can influence cell type-specific susceptibility to induction of epimutations by exposure to BPS, and that different combinations of these variables ultimately determine the quantity and quality of epimutations induced by exposure to BPS.

Endogenous fetal germline cells (e.g. PGCs and prospermatogonia in males) normally display lower levels of global DNA methylation and higher levels of chromatin accessibility than pluripotent or somatic cell types (Hajkova, 2011 [↗](#)), yet PGCLCs developed fewer epimutations following exposure to BPS than the other cell types. This appears to be explained by the fact that germline cells do not express either estrogen receptor whereas pluripotent and somatic cell types express one or both of the two estrogen receptors, apparently rendering them more susceptible to induction of epimutations following exposure to the estrogen mimetic, BPS. This is consistent with the notion that EDCs impose disruptive effects by interfering with canonical endocrine signaling

pathways, and that notion is further supported by our observation that, in the cell types that express one or both estrogen receptors, we observed a prevalence of BPS-induced epimutations in genomic regions containing nearby EREs, whereas in PGCLCs, which do not express either estrogen receptor, we did not observe a correlation between the locations of BPS-induced epimutations and genomic EREs.

Nevertheless, we did observe induction of epimutations in PGCLCs exposed to BPS, suggesting that while cell types expressing relevant endocrine receptors may display elevated susceptibility to EDC-induced epimutagenesis presumably based on disruption of canonical endocrine signaling, this is not the only mechanism by which EDCs can induce epimutations. Indeed, our results support the suggestion that there may be at least two types of epimutations induced by exposure to EDCs – one that is independent of canonical endocrine signaling and so occurs in all cell types regardless of expression of relevant endocrine receptors or the genomic location of relevant HREs, and another that does involve disruption of canonical endocrine signaling and so is elevated in cell types expressing relevant endocrine receptors and predisposes the induction of epimutations in genomic regions near relevant HREs. Beyond this, a comparison of the quantities of BPS-induced epimutations in pluripotent and somatic cell types, which both express one or both estrogen receptors, but which differ with respect to elevated genome-wide chromatin accessibility in pluripotent cells compared to much more limited chromatin accessibility in somatic cell types (Cantone and Fisher, 2013 [DOI](#); Santos et al., 2002 [DOI](#)), suggests that chromatin accessibility also contributes to susceptibility to EDC-induced epimutagenesis.

One of our most intriguing findings was that, in addition to estrogen-receptor expressing somatic and pluripotent cell types developing more epimutations than non-estrogen-receptor expressing germ cells in response to exposure to BPS, the occurrence of epimutations in the former (somatic and pluripotent) cell types was predominantly in enhancer regions whereas that in the latter (germ) cell type was predominantly in gene promoters. As noted above, there was a significant increase in the occurrence of BPS-induced epimutations near EREs in somatic and pluripotent cell types compared to germ cells. A genome wide analysis confirmed that ERE half-sites occur more frequently in enhancer regions than in promoter regions (Supplementary file 1b) supporting our suggestion that the higher prevalence of BPS-induced epimutations we observed in enhancer regions in somatic and pluripotent cell types was due to disruption of canonical hormone signaling.

Our finding that PGCLCs, which do not express endocrine receptors, still accrued epimutations following exposure to BPS indicates that in addition to inducing epimutations by disrupting canonical endocrine signaling pathways, exposure to EDCs can induce epimutations via mechanisms that do not involve disruption of such pathways. This notion is further supported by the occurrence of BPS-induced epimutations in genomic regions that do not include nearby EREs, even though a prevalence of the epimutations induced in PGCLCs were located in gene promoters. Pathway analysis of the genes in which promoter-region epimutations were induced by exposure of PGCLCs to BPS suggests an apparent endocrine-signaling independent group of genes that can become disrupted by exposure to an EDC via non-canonical signaling which may occur through the PI3K/AKT pathway via G-protein coupled receptors. While we observed expression of a number of G-protein coupled receptors known to bind to either 17 β -estradiol or BPA, further studies will be required to determine if any of these contribute to susceptibility to BPS epimutagenesis.

As expected, disruption of normal epigenetic programming induced by exposure of cells to BPS also led to dysregulation of gene expression. Thus, RNA-seq analysis detected differentially expressed genes in all cell types exposed to BPS. Surprisingly, the greatest number of DEGs was detected in PGCLCs which was the cell type that developed the lowest overall number of epimutations in response to exposure to BPS. However, PGCLCs showed a higher prevalence of BPS-induced epimutations in gene promoter regions than was observed in any of the other cell

types, so it appears that promoter-region epimutations impose the most direct impact on gene expression. As noted above, fetal germ cells normally display the lowest level of global DNA methylation of any cell type at any developmental stage. This is characteristic of the epigenetic ground state that accrues uniquely in developing germ cells ([Hajkova, 2011](#)), and is believed to reflect a generally more accessible chromatin state genome-wide in germ cells than in other cell types. Thus, it may be that the epigenetic ground state in developing germ cells renders this cell type uniquely susceptible to epimutagenesis in gene promoter regions, thereby predisposing a high level of dysregulation of gene expression in response to exposure to an EDC.

Finally, our *in vitro* system allowed us to follow the fate of BPS-induced epimutations through a major transition in cell fate from pluripotent to germline cells during which a portion of the normal generational germline epigenetic reprogramming process is known to occur ([Kurimoto and Saitou, 2018](#)). Thus, when we induced iPSCs previously exposed to 1 μ M BPS to differentiate into PGCLCs, we found that a substantial prevalence of epimutations persisted during this transition. However, very few of the specific DMCs or the associated DEGs detected in the BPS-exposed iPSCs persisted in the subsequently derived PGCLCs. Specifically, only 3.7 % of DMCs and 8.4 % of DEGs were conserved between the BPS-exposed iPSCs and the subsequently derived PGCLCs. This suggests that exposure of cells to EDCs has the potential to disrupt epigenetic programming in a way that then interferes with subsequent generational reprogramming that normally accompanies either germline to pluripotent or pluripotent to germline transitions in cell fate. Recent *in vivo* studies have suggested that exposure of gestating female mice to the EDC tributyltin results in disruption not simply of the pattern of epigenetic modifications but also of the underlying chromatin landscape ([Chamorro-García et al., 2021](#); [Chang et al., 2022](#)). In this context, our results are consistent with the suggestion that exposure of cells to EDCs can disrupt the underlying chromatin landscape (e.g. the pattern of A and B chromatin compartments) such that when the “normal” reprogramming machinery then acts on this disrupted landscape it corrects many of the originally induced epimutations, but simultaneously induces many novel epimutations *de novo*. This would predispose ongoing abnormalities in the underlying chromatin landscape that would, in turn, lead to the recurring correction of many existing epimutations in concert with the genesis of many novel epimutations during each subsequent generation.

Taken together, our results demonstrate the utility of an *in vitro* cell culture approach for pursuing molecular mechanisms underlying environmentally induced disruption of normal epigenetic programming manifest as the formation of epimutations and dysregulated gene expression. This approach clearly affords unique potential to reveal mechanisms responsible for the initial induction of epimutations in response to direct exposure of cells to disruptive effects such as EDCs, as well as offering potential means to elucidate mechanisms by which environmentally induced epimutations are then propagated within the exposed individual and then to that individual’s descendants. Knowledge of these mechanisms will afford the best opportunity to understand how these defects may, in the future, be better diagnosed, treated, and/or prevented. With the ever-expanding catalogue of potentially hazardous man-made compounds permeating our environment, it is increasingly important that we recognize the potential dangers such compounds may impose and maximize our ability to protect ourselves from those dangers.

Materials and methods

Key resources table

		Scientific		
Chemical compound	B-27™	Thermo Fisher Scientific	12587010	(50X), minus vitamin A
Chemical compound	Penicillin-streptomycin	Thermo Fisher Scientific	15070063	(5,000 U/mL)
Chemical compound	GlutaMAX™	Thermo Fisher Scientific	35050061	(100X)
Chemical compound	2-Mercaptoethanol	Thermo Fisher Scientific	21985023	(1000X)
Chemical compound	CHIR99021	BioVision	1677-5	
Chemical compound	PD0325901	Amsbio	04-0006-02	10 mM in DMSO
Chemical compound	Recombinant Human/Murine/Rat Activin A	PeproTech	120-14	Insect derived
Chemical compound	Recombinant Human FGF-Basic (FGF-2/bFGF)	Thermo Fisher Scientific	13256-029	
Chemical compound	KnockOut™ Serum	Thermo Fisher Scientific	10828028	
Chemical compound	Glasgow's MEM (GMEM)	Thermo Fisher Scientific	11710035	
Chemical compound	Recombinant human bone morphogenetic protein 4 (BMP-4)	R&D Systems	314-BP-010	
Chemical compound	Recombinant mouse stem cell factor (SCF)	R&D Systems	455-MC-010	
Chemical compound	Recombinant human epidermal growth factor (EGF), carrier free (CF)	R&D Systems	2028-EG-200	
Chemical compound	Dulbecco's phosphate-buffered saline (DPBS)	Thermo Fisher Scientific	14040133	
Chemical compound	Deoxyribonuclease I (DNaseI)	Millipore Sigma	DN25	From bovine pancreas
Chemical compound	Trypsin (2.5%)	Thermo Fisher Scientific	15090046	No phenol red
Chemical	Soybean trypsin	Thermo	17075029	

compound	inhibitor	Fisher Scientific		
Chemical compound	Collagenase type IV	Worthington	LS004188	From <i>Clostridium histolyticum</i>
Chemical compound	Sertoli Cell Medium	ScienCell Research Laboratories	4521	
Chemical compound	Ethanol (EtOH)	Fisher	BP28184	(200 Proof)
Chemical compound	BSA	Millipore Sigma	A9085	
Chemical compound	Heat inactivated (HI) FBS	Thermo Fisher Scientific	10082147	
Chemical compound	Trypsin-EDTA (0.25%)	Thermo Fisher Scientific	25200072	With phenol red
Chemical compound	Bisphenol S (BPS)	Millipore Sigma	43034-100MG	
Chemical compound	Phenol:Chloroform:Iso amyl Alcohol (25:24:1, v/v)	Thermo Fisher Scientific	15593031	
Chemical compound	TRIzol™	Thermo Fisher Scientific	15596026	
Chemical compound	Isopropanol	Thermo Fisher Scientific	327272500	
Chemical compound	Proteinase K Solution (20 mg/mL)	Thermo Fisher Scientific	25530049	
Chemical compound	MaXtract High Density	Quiagen	129046	Phase lock gel tubes
Chemical compound	Sodium Acetate Solution	Thermo Fisher Scientific	R1181	3M, pH 5.2
Chemical compound	Glycogen (5 mg/ml)	Thermo Fisher Scientific	AM9510	
Chemical compound	NaCl	Thermo Fisher Scientific	J21618.36	
Chemical compound	Tris base	Millipore Sigma	77-86-1	
Chemical compound	Ethylenediaminetetra acetic acid (EDTA)	Millipore Sigma	E9884-100G	
Chemical	Sodium dodecyl	Millipore	151-21-3	

compound	sulfate (SDS)	Sigma		
Chemical compound	Triton X-100	Thermo Fisher Scientific	85111	
Chemical compound	RQ1 DNase	Promega	M6101	
Chemical compound	Propidium iodide	BioLegend	421301	FCy 5 $\mu\text{L}/10^6$ cells
Antibody	ER α	Thermo Fisher Scientific	MA1-310	Host: mouse, ICC 1:100
Antibody	ER β	GeneTex	GTX70174	Host: mouse, ICC 1:100
Antibody	INHA	Invitrogen	PA5-13681	Host: rabbit, ICC 1:25
Antibody	FSHR	Affinity	AF5477	Host: rabbit, ICC 1:250
Antibody	SOX9	Abcam	ab185966	Host: rabbit, ICC 1:100
Antibody	GAPDH	Novus	NB300-221	Host: mouse, ICC 1:100
Antibody	WT1	Novus	NBP2-67587	Host: rabbit, ICC 1:100
Antibody	FUT4	GeneTex	GTX34467	Host: rabbit, ICC 1:50
Antibody	NANOG	Abcam	ab80892	Host: rabbit, ICC 1:100
Antibody	POU5F1	Abcam	ab19857	Host: rabbit, ICC 1:200
Antibody	SOX2	Abcam	ab97959	Host: rabbit, ICC 1:200
Antibody	ID4	Thermo Fisher Scientific	PA5-26976	Host: rabbit, ICC 1:50
Antibody	Goat Anti Mouse Alexa 488	Abcam		Host: goat, ICC 1:1000
Antibody	Goat Anti Rabbit Alexa 647	Abcam	ab150199	Host: goat, ICC 1:200
Antibody	FUT4 (IgM, κ), brilliant violet 421	BioLegend	125614	Host: mouse, FCy 5 $\mu\text{L}/10^6$ cells
Antibody	ITGB3 (IgG), PE	BioLegend	104307	Host: hamster, FCy 1 $\mu\text{L}/10^6$ cells
Antibody	IgM, κ Isotype control, brilliant violet 421	BioLegend	562704	Host: mouse, FCy 1.25 $\mu\text{L}/10^6$ cells
Antibody	IgG Isotype control, PE	BioLegend	400907	Host: hamster, FCy 1 $\mu\text{L}/10^6$ cells
Commercial	RNA Clean &	Zymo	R1016	

assay or kit	Concentrator-5	Research		
Commercial assay or kit	Genomic DNA Clean & Concentrator-10	Zymo Research	D4011	
Commercial assay or kit	EZ DNA Methylation Kit	Zymo Research	D5001	
Commercial assay or kit	SuperScript™ III One-Step RT-PCR System with Platinum™ Taq DNA Polymerase	Thermo Fisher Scientific	12574026	
Commercial assay or kit	PowerTrack™ SYBR Green Master Mix for qPCR	Thermo Fisher Scientific	A46109	
Commercial assay or kit	Infinium Mouse Methylation BeadChip	Illumina	20041558	
Commercial assay or kit	RNA ScreenTape & Reagents	Agilent	5067-5576	
Commercial assay or kit	DNA ScreenTape & Reagents	Agilent	5067-5583	
Commercial assay or kit	Qubit dsDNA (Broad Range) BR Assay Kit	Thermo Fisher Scientific	Q32850	
Commercial assay or kit	Qubit RNA (high sensitivity) HS Assay Kit	Thermo Fisher Scientific	Q32855	
Commercial assay or kit	NEBNext® Ultra™ II Directional RNA Library Prep Kit for Illumina®	New England BioLabs	E7765	
Commercial assay or kit	NEBNext® Poly(A) mRNA Magnetic Isolation Module	New England BioLabs	E3370	
Software, algorithm	ZEISS ZEN Microscopy Software	https://www.zeiss.com/microscopy/en/products/software/zeiss-zen.html	ZEN 3.7	
Software, algorithm	Primer-BLAST	https://www.ncbi.nlm.nih.gov/tools/primer-blast/		
Software, algorithm	QuantSudtio™ Design & Analysis Software	https://www.thermofisher.com/us/en/home/technical-resources/so	QuantSudtio™ v1.5.1	

		ftware-downloads/quantstudio-3-5-real-time-pcr-systems.html		
Software, algorithm	Fiji	https://fiji.sc/	Fiji v1.54f	(Schindelin et al., 2012)
Software, algorithm	Bfastq2	https://support.illumina.com/downloads/bcl2fastq-conversion-software-v2-20.html	Bcl2fastq2 v2.20	
Software, algorithm	FastQC	https://www.bioinformatics.babraham.ac.uk/projects/fastqc/	FastQC 0.12.0	(Andrews et al., 2023; Smith and de Sena Brandine, 2021)
Software, algorithm	Wg-blimp	https://github.com/MarWoes/wg-blimp	Wg-blimp v0.10.0	(Lehle and McCarrey, 2023; Wöste et al., 2020)
Software, algorithm	R-Project for Statistical Computing	http://www.r-project.org/	R 4.2.1	Packages: Sesame (Ding et al., 2023; Triche et al., 2013; Zhou et al., 2022, 2018), stringr (Wickham and RStudio, 2022), kintr (Xie et al., 2023), SummarizedExperiment (Morgan et al., 2023), ggrepel (Slowikowski et al., 2023), pals (Wright, 2023), wheatmap (Zhou, 2022), tidyr (Wickham et al., 2023) magrittr (Bache et al., 2022), dplyr (Wickham et al., 2023), ggplot2 (Wickham et al., 2023),

				ggvenn (Yan, 2023), RColorBrewer (Neuwirth, 2022), Rideogram (Hao et al., 2020), AnnotationDbi (Pagès et al., 2023), Mus.musculus (Team BC, 2015), BSgenome.Mmusculus.UCS C.mm10 (Team TBD, 2021), GenomicRanges (Lawrence et al., 2013), universalmotif (Tremblay, 2023), memes (Nystrom, 2023), Plyranges (Lee et al., 2019), rtracklayer (Lawrence et al., 2009), Rsubread (Liao et al., 2019), edgeR (Chen et al., 2016; McCarthy et al., 2012; Robinson et al., 2010)
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Animal procedures

Mice were used as the primary source of cells for the establishment of cultures for the analysis of differential cell type-specific susceptibility to exposure to the EDC BPS at the cellular level. All mice were euthanized prior to dissection and isolation of the tissue or cell type to be used for experiments. All animals were maintained in one of the UTSA on-campus vivaria and housed under controlled temperature and humidity conditions in a fixed 12-hour light, 12-hour dark cycle with free access to food and water. Euthanasia was performed by trained personnel using continuous CO₂ exposure at a rate of 3L/min until one minute after breathing had stopped. Euthanasia was confirmed by cervical dislocation. Tissue from euthanized mice was used to obtain somatic (Sertoli, granulosa, and mouse embryonic fibroblast [MEF]) cells, and primordial germ cells [PGCs]. Both Sertoli and granulosa cells were isolated from respective male and female mice euthanized at postnatal day 20 (P20). MEFs and PGCs were isolated from fetuses at embryonic day 13.5 postcoitum (E13.5).

In vitro generation and/or culture of pluripotent, somatic, and germ cells

Pluripotent cell culture

Male mouse iPSCs were derived from a transgenic mouse line carrying the *tetO-4F2A* cassette obtained from The Jackson Laboratories (011011) (The Jackson Laboratory, ME USA). iPSCs can be induced from essentially any cell type carrying this reprogrammable cassette by exposure to Doxycycline for one week as previously described (Carey et al., 2010 [DOI](#); Hochedlinger et al., 2005 [DOI](#)). For this project, iPSCs were reprogrammed from MEFs isolated from a single male (XY) E13.5 mouse fetus.

Reprogrammed colonies were picked and expanded via sub-passaging. Validation of reprogrammed pluripotency was determined by positive immunocytochemical (ICC) staining for the pluripotent markers POU5F1, SOX2, NANOG, and FUT4 (Supplementary file 1c and 1f). Karyotyping was done on each candidate iPSC line produced to confirm normal chromosome number and XY sex chromosome constitution of each line prior to aliquots being prepared for

long-term storage in liquid nitrogen (Supplementary file 1d). Only male (XY) iPSC lines were used for this project. Upon thawing, iPSCs were initially maintained on CF-1 feeder cells for a minimum of three passages. Cells were cultured in DMEM supplemented with 15% fetal bovine serum (FBS) and 1000 U/mL leukocyte inhibitory factor (LIF). Once pluripotent cells were stabilized in culture, they were transitioned to feeder-free conditions and were cultured in N2B27 media supplemented with 2i and LIF for two-three passages prior to use in chemical exposure experiments. A complete list of the media components along with catalog numbers can be found in Supplementary file 1e.

Sertoli cell culture

Primary cultures of Sertoli cells were established as previously described (Karl and Griswold, 1990 [DOI](#)). Briefly, whole testes were dissected from juvenile mice at P20. Following removal of the tunica albuginea from each individual testis, the bundles of seminiferous tubules were physically chopped using a sterile razor blade to break down the coiled structure of tubules increasing the surface area and rendering them easier to digest. These shortened fragments of seminiferous tubules were then digested in a mixture of 2.5% trypsin and 6.64 mg/ml DNaseI in DPBS for 25 minutes at 37°C. After these enzymes were inactivated, the tubule sections were washed multiple times, then treated with a final enzymatic mixture of collagenase IV (0.70 mg/mL) and DNaseI (6.64 mg/mL) for 10 minutes to permeabilize the thick collagen layer on the exterior of the seminiferous tubules to allow the Sertoli cells to migrate out away from the tubules in culture. After checking the digested tubules under a microscope to confirm the collagen layer had been permeabilized, the digested tubules were spun down to wash away the enzymes, and then plated in Sertoli cell media containing retinoic acid from ScienCell™ (4521) (ScienCell™ Research Laboratories, CA USA). To remove contaminating germ cells from this primary culture, the cells were treated with hypotonic shock treatment on the morning of the second day of culture, and the enriched Sertoli cells were washed and allowed to recover with fresh media for at least two hours. The enriched primary cultures of Sertoli cells were then ready to be used for chemical exposure experiments. The estimated purity of the culture was > 80% based on ICC staining for the Sertoli cell marker SOX9 and WT1 (Supplementary file 1f). A full procedure for the establishment of primary cultures of Sertoli cells can be found in Supplementary file 1g.

Granulosa cell culture

Primary cultures of granulosa cells were established by selecting early preantral primary follicles from enzymatically digested ovary tissue as described previously (Monti and Redi, 2016 [DOI](#); Roy and Greenwald, 1996 [DOI](#)), followed by breaking selected primary follicles down into a single cell suspension that could be plated and maintained. To isolate preantral primary follicles we dissected ovaries from female mice at P20 and then enzymatically digested the ovary using collagenase IV (560 U/pair of ovaries) for 25 minutes at 37°C with constant agitation. This digestion was stopped by addition of buffer with 0.5% BSA and the mixture was spun down at 60xg for 5 minutes to pellet the cells. The cells were resuspended in PBS with 0.5% BSA and transferred to a sterile petri dish under a stereomicroscope.

Primary follicles were individually picked from the solution using a glass needle with suction control and moved into a clean droplet of PBS containing 0.5% BSA. Selected primary follicles were then spun down at 1000xg for 5 mins. After spinning, the supernatant was removed, and the cells were finger-vortexed to resuspend the pellet. The follicles were then digested to single cell suspension by addition of pre-warmed (37°C) 0.25% trypsin and incubated for 5 mins with regular pipetting followed by pelleting again at 1000xg for 5 mins. The trypsin-containing supernatant was removed, and the granulosa cells were resuspended in DMEM supplemented with 15% FBS and plated for maintenance in culture. The oocytes were non-adherent and were washed away on day 2 when changing the media. The estimated purity of the culture was > 90% based on ICC staining for the granulosa cell marker FSHR and INHA (Supplementary file 1f). The granulosa cells were then ready to be used for EDC exposure experiments. A full procedure for the establishment of primary cultures of granulosa cells can be found in Supplementary file 1h.

Primordial germ cell like cell culture

Mouse iPSCs were differentiated into PGCLCs as previously described (Hayashi et al., 2011 [↗](#)). Briefly, iPSCs maintained in N2B27 supplemented with 2i and LIF under feeder free conditions were differentiated to EpiLCs for two days by the addition of activin A and basic fibroblast growth factor (bFGF) to the N2B27 media. After two days the EpiLC intermediate cells were passaged to low adherence round bottom plates for 4 days in GK15 media containing bone morphogenic protein 4 (BMP4), stem cell factor (SCF), epidermal growth factor (EGF), and LIF to induce the differentiation of a subset of cells (2-3%) in the resulting aggregate to become PGCLCs. During this four-day period, cell aggregates were ready for EDC exposure (see below). Following EDC exposure, cell aggregates were removed from the low adherence round bottom plates, dissociated into a single cell suspension, and PGCLCs were fluorescence-activated cell sorted (FACS) on a BD FACSaria™ II in the UTSA Cell Analysis Core to recover cells that were double-positive for FUT4 and ITGB3. A full list of the media components and catalog numbers for the differentiation of iPSCs to PGCLCs along with our FACS gating for double positive FUT/ITGB3 PGCLCs can be found in Supplementary file 1e and 1i, respectively. Additional data demonstrating validation of PGCLCs by qRT-PCR and ICC staining are shown in Supplementary file 1j.

Immunocytochemistry (ICC)

Cells were immunolabeled as previously described (Rodig, 2022 [↗](#)) to validate the purity of cell primary cultures using known cell specific markers (Supplementary file 1f) detect the presence or absence of relevant endocrine receptors ER α and ER β at the protein level. Pluripotent (iPS) and somatic (Sertoli and granulosa) cells were grown on 13mm Thermanox® plastic coverslips (174950) prior to fixation, permeabilization, and immunolabeling (Nalge Nunc International, NY USA). Non-adherent germ (PGC and PGCLC) cells were spun down onto poly-L-lysine coated slides (63410-01) at 400 RPM for one minute using a Thermo Shandon CytoSpin III Cytocentrifuge from Rakin (Electron Microscopy Sciences, PA USA & Rakin Biomedical Corporation, MI USA). Cells were fixed with 4% formaldehyde for 10 mins at room temp (RT) and then washed three times for five minutes each with ICC buffer (PBS containing 0.01% Triton X-100 detergent) to permeabilize the cell and nuclear membranes prior to blocking with 5% goat serum which was added to the ICC buffer and incubated for one hour at RT. Following blocking, primary antibodies, in ICC buffer, were added to the slides and left to incubate overnight at 4°C. The following day, slides were washed three times with ICC buffer, and fluorescent secondary antibodies in ICC buffer were then added to the slides and left to incubate in the dark for 1 hour. After secondary antibody incubation, cellular nuclei were stained with DAPI at a 1:1000 dilution in ICC buffer for 7 minutes, followed by 3 final washes with ICC buffer for 5 minutes. Coverslips then were transferred to microscope slides and mounted with 5-10 μ L of VECTASHEILD Antifade Mounting Medium (H-1000) and sealed with clear nail polish before being imaged on a Zeiss AXIO Imager.M1 Fluorescence Microscope (Vector Laboratories Inc, CA USA & Zeiss Group, Oberkochen DE). Images were processed for contrast and brightness enhancement and for the addition of scale bars using Fiji (Schindelin et al., 2012 [↗](#)).

Information about the primary and secondary antibodies along with catalogue numbers and dilutions used can be found in the Key resources table.

Quantitative RT-PCR (qRT-PCR)

Total RNA was extracted from cells as described previously (Rio et al., 2010 [↗](#)) and treated with 1.5 U/ μ g total RNA RQ1 DNase1 (M6101) to remove contaminating genomic DNA (Promega™ Corporation, WI USA). 50 ng of cleaned RNA was retrotranscribed with SuperScript™ III as recommended by the manufacturer (Invitrogen, MA USA). Primers were designed using Primer-BLAST (Ye et al., 2012 [↗](#)) from NCBI. A complete list of all primer sequences used in this study can be found in Supplementary file 1j.

Relative expression levels of selected genes were assessed by real-time PCR using the PowerTrack™ SYBR Green Master Mix according to the manufacturer's instructions (Applied Biosystems, MA USA) then run on a QuantStudio™ 5 Real-Time PCR System (Thermo Fisher Scientific, MA USA) and analyzed with QuantStudio™ Design & Analysis Software. Each sample was normalized based on constitutive expression of the *Gusb* reference gene to obtain the ΔCt [(2⁻[Ct_{gene}-Ct_{Gusb}])].

BPS exposure design

Despite numerous studies illustrating the dangers of estrogenic mimetic EDCs, the EPA has currently not published any limits concerning the maximum acceptable dose of BPS below which exposure on a daily basis is considered to be safe. Thus for this study, we established our initial testing range of 1 μ M, 50 μ M, and 100 μ M of BPS based on the limit established for BPA at ≤ 4.44 μ M (Ribeiro et al., 2019 [↗](#)) selecting 1 dose below the established safe limit and two doses that exceed that limit. For all cell types except PGCLCs, BPS was added to media gassed with 5% CO₂, 5% O₂, and balanced N₂ and then injected into sealed T-25 cell culture flasks and left to incubate for 24 hours (Supplementary file 1k). The media was gassed to ensure enrichment to 5% CO₂, which is normally regulated by the cell incubator, limiting the potential for the pH to change in the media during this exposure period due to the absorption of CO₂ by incubating cells. Following 24 hours of exposure, media containing BPS was removed and cells were washed and allowed to recover in fresh untreated media (without BPS or EtOH) for 24 hours prior to harvesting. PGCLCs were not suitable for this exposure paradigm as EpiLCs undergo differentiation to PGCLCs in cell aggregates formed in low-adherence round-bottom 96 well plates and cannot be maintained in T-25 sealed flasks. Therefore, diluted BPS was added to PGCLC media and added to cells in round-bottom 96 well plates to incubate for 24 hours prior to a shortened wash and “chase” period of 8 hours prior to cell sorting.

Methylation beadchip analysis

A total of 1 μ g of extracted genomic DNA was bisulfite-converted with the EZ DNA Methylation Kit (D5001) and modified according to the manufacturer's recommendations (Zymo Research, CA USA & Illumina, CA USA). These samples were run on the Infinium Mouse Methylation BeadChip Array following the Illumina Infinium HD Methylation protocol. This beadchip array includes 297,415 cytosine positions within the mouse genome (CpG sites, non-CpG sites, and random SNPs). The methylation score for each CpG is represented as a β -value which is a ratio of the fluorescence intensity ranging between 0 (unmethylated) and 1 (methylated). Arrays were scanned by HiScan (Illumina, CA USA). Quality control (Supplementary file 1m) and downstream data processing of the .idat files was using the Sensible Step-wise Analysis of DNA Methylation BeadChips (SeSAMe) Bioconductor package (Ding et al., 2023 [↗](#); Triche et al., 2013 [↗](#); Zhou et al., 2022 [↗](#), 2018 [↗](#)). DNA methylation levels of differentially methylated cytosines (DMCs) are determined using mixed linear models. This general supervised learning framework identifies CpG loci whose differential methylation is associated with known control vs. treated co-variables. CpG probes on the array were defined as having differential changes that met both p-value and FDR ≤ 0.05 significant thresholds between treatment and control samples for each cell type analyzed.

Additionally, we followed up our DNA methylation analysis of individual dCpGs by identifying differentially methylated regions (DMRs). DMRs were created by grouping all CpGs measured on the array into clusters using Euclidean distance (Ding et al., 2023 [↗](#); Triche et al., 2013 [↗](#); Zhou et al., 2022 [↗](#), 2018 [↗](#)). The p-values from the differential methylation of individual CpGs within the resulting CpG clusters were aggregated, and clusters were then filtered selecting for regions that contained a p-value ≤ 0.05 (Ding et al., 2023 [↗](#); Triche et al., 2013 [↗](#); Zhou et al., 2022 [↗](#), 2018 [↗](#)).

RNA-seq

Total RNA was extracted from cells using Trizol as previously described (Rio et al., 2010). Contaminating genomic DNA was removed by RQ1 DNase (M6101) treatment according to the manufacturer's instructions (Promega Corporation, WI USA). RNA concentration was determined using Qubit (Q32855) and RNA integrity (RIN) scores were determined using tape station (5067-5576) according to the manufacturer's instructions (Thermo Fisher Scientific Inc, MA USA & Agilent Technologies, Inc CA USA). Strand-specific RNA-seq libraries were prepared with the NEBNext® Ultra™ II Directional RNA Library Prep Kit for Illumina® sequencing (E7760S) according to the manufacturer's protocol (New England Biolabs, MA USA). Briefly, this process consisted of poly(A) RNA selection, RNA fragmentation, and double stranded cDNA generation using random oligo(dT) priming followed by end repair to generate blunt ends, adaptor ligation, strand selection, and polymerase chain reaction amplification to generate the final library. Distinct index adaptors were used for multiplexing samples across multiple sequencing lanes. Sequencing was performed on an Illumina NovaSeq 6000 instrument yielding sequences of paired end 2x50 base pair runs. Demultiplexing was performed with the Illumina Bcl2fastq2 program (Illumina, CA USA).

RNA-seq expression analysis

The quality of the fastq reads was checked using FastQC (Andrews et al., 2023; Smith and de Sena Brandine, 2021) (Supplementary file 1n). Reads were aligned to the mm10 mouse reference genome using the Rsubreads package to produce read counts (Liao et al., 2019). These were then used for differential gene expression analysis using the edgeR package (Chen et al., 2016; McCarthy et al., 2012; Robinson et al., 2010). Briefly, gene counts were normalized using the trimmed mean of M-value normalization (TMM) method before determining counts per million (CPM) values (Robinson and Oshlack, 2010). For a gene to be classified as showing differential gene expression between BPS-treated and EtOH vehicle-only control samples, a threshold of both a Benjamini-Hochberg adjusted p-value ≤ 0.05 and a false discovery rate (FDR) ≤ 0.05 had to be met.

EM-Seq

Genomic DNA was extracted from cells as previously described (Sambrook and Russell, 2006). Smaller fragmented DNA (≤ 10 kb) and contaminating RNA were removed by cleaning the genomic DNA on a genomic DNA clean and concentrator™-10 column (D4011), according to the manufacturer's instructions (Zymo Research, CA USA). DNA concentration was determined using a Qubit (Q32850) and genomic DNA (100 ng) was sheered using a Bioruptor® and the size of the sheered DNA was determined using a TapeStation 4200 according to the manufacturer's instructions in the UTSA Genomics Core (Diagenode Inc, NJ USA & Agilent Technologies, Inc CA USA). Large size (470-520 bp) EM-seq libraries were prepared with the NEBNext® Enzymatic Methyl-seq kit (E7120S) according to the manufacturer's protocol (New England Biolabs, MA USA). Briefly, this process consisted of A-tailing, adaptor ligation, DNA oxidation by TET2 initiated by the addition of Fe (II), strand denaturation with formamide, deamination by APOBEC3A, polymerase chain reaction amplification, and bead selection to generate the final libraries. Distinct index adaptors were used for multiplexing samples across multiple sequencing lanes. Sequencing was performed on an Illumina NovaSeq 6000 instrument yielding sequences of paired end 2x150 base pair runs. Demultiplexing was performed with the Illumina Bcl2fastq2 program (Illumina, CA USA).

EM-Seq analysis of genome-wide DNA methylation patterns

EM-seq data was processed using the comprehensive wg-blimp v10.0.0 software pipeline (Lehle and McCarrey, 2023; Wöste et al., 2020). In brief, reads were trimmed prior to initiating the wg-blimp pipeline using Trim Galore (https://www.bioinformatics.babraham.ac.uk/projects/trim_galore/). Sequenced reads were aligned to the mm10 genome using gemBS (King et al., 2020);

Merkel et al., 2019 [↗](#); Schilbert et al., 2020 [↗](#)). The BAM files from alignment underwent a series of QC tests including read quality scoring by FastQC (Andrews et al., 2023 [↗](#)), overall and per-chromosome read coverage calculation, GC content, duplication rate, clipping profiles by Qualimap (Okonechnikov et al., 2016 [↗](#)), and deduplication by the Picard toolkit (<https://broadinstitute.github.io/picard/> [↗](#)) (Supplementary file 1o).

Methylation calling was performed by MethylDackel (<https://github.com/dpryan79/MethylDackel> [↗](#)) and statistically significant DMC/DMR calling was performed by the metilene (Jühling et al., 2016 [↗](#)) and BSmooth (Hansen et al., 2012 [↗](#)) algorithms. Metilene uses a binary segmentation algorithm combined with a two-dimensional statistical test that allows the detection of DMCs/DMRs in large methylation experiments with multiple groups of samples. BSmooth uses a local-likelihood approach to estimate a sample-specific methylation profile, then computes estimates of the mean differences and standard errors for each CpG to form a statistic similar to that used in a t-test. Finally, potential regulatory regions were identified through the use of MethylSeekR (Burger et al., 2013 [↗](#)). Results from the pipeline were displayed in the wg-blimp interactive results web browser that was built using the R Shiny local browser hosting framework (Chang et al., 2023 [↗](#)).

Data availability

Methylation beadchip, EM-seq, and RNA-seq data discussed in this publication can be obtained from the SRA BioProject (PRJNA1026145).

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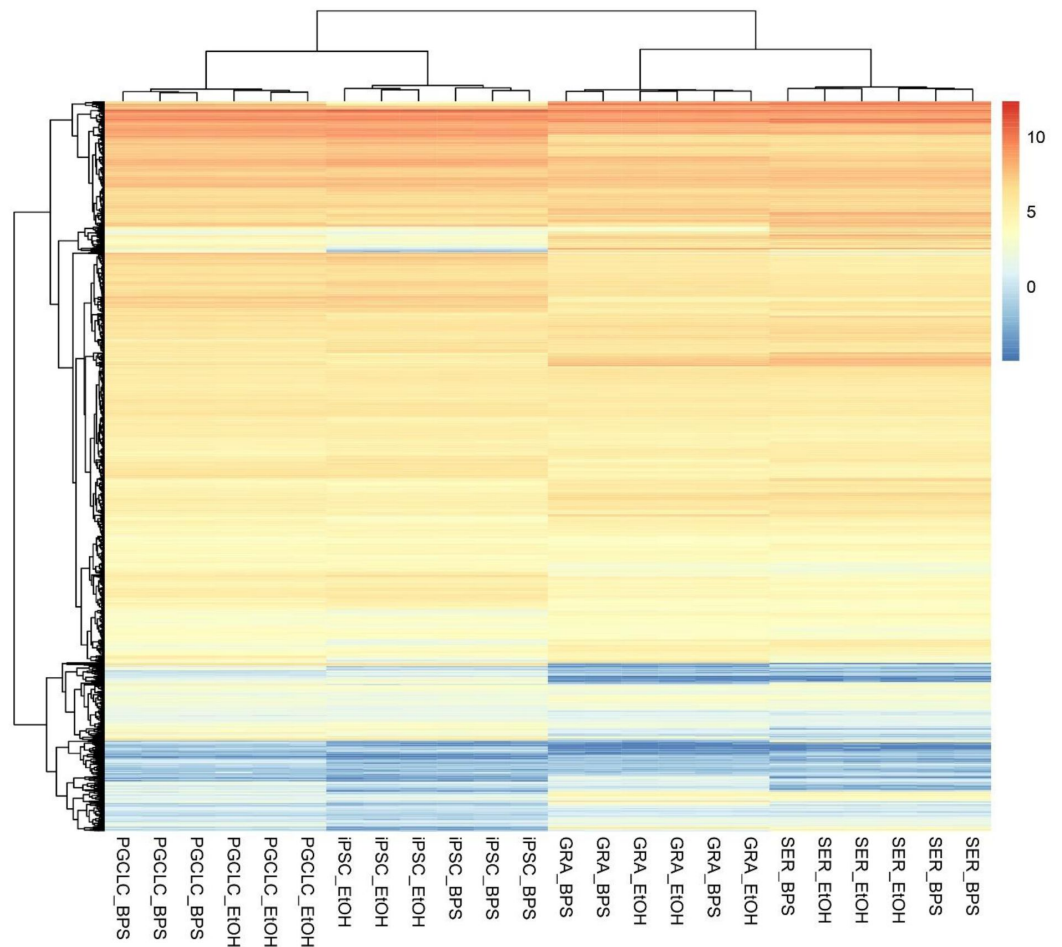
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Ethics

All experimental procedures involving animals were approved in advance by the UTSA IACUC (Protocols MU015 and MU016).

Supplemental files

Supplementary file 1a

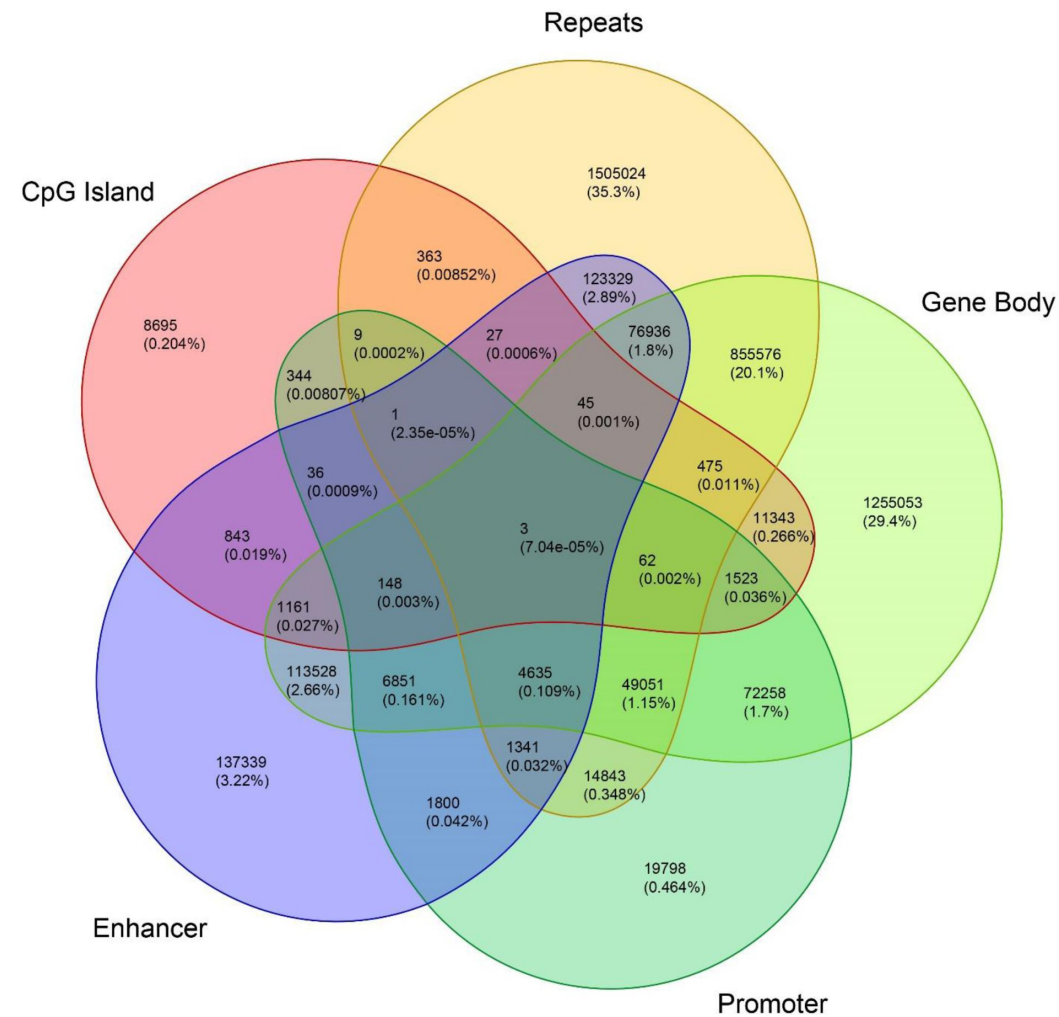


Supplement figure 1.

Differential expression of potential endocrine-signaling independent DMCs.

Heatmap displaying the relative expression of genes with promoters enriched for apparent endocrine-signaling independent DMCs. The majority of these genes are actively expressed in all cell types and an overall largely similar pattern of expression across all cell types.

Supplementary file 1b



Supplement figure 2.

Genome-wide annotation of ERE half-sites.

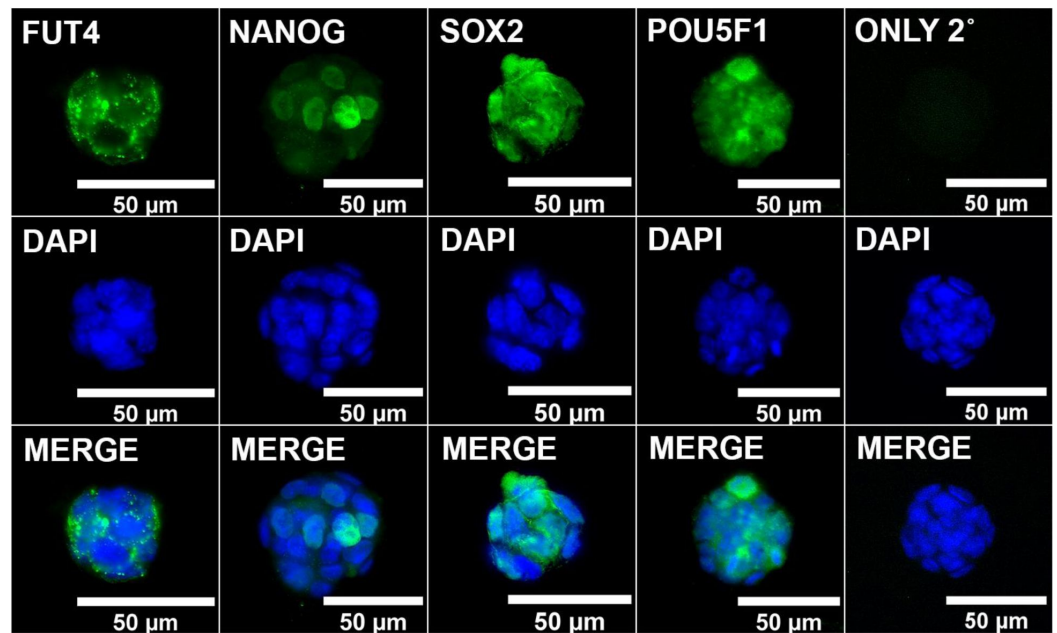
Venn diagram displaying the identification of ERE half-sites localized in known genic and intergenic regions.

CpG Islands	Repeat Regions	Gene Bodies	Promoters	Enhancers
25,079	2,631,743	2,448,668	172,707	468,072

Supplement table 1.

Summary of ERE annotations

Supplementary file 1c



Supplement figure 3.

ICC validation of MF5-9-1 iPSCs.

iPSCs from reprogrammed MEFs were validated for immunolabeling with known pluripotency markers along with negative control.

Supplementary file 1d

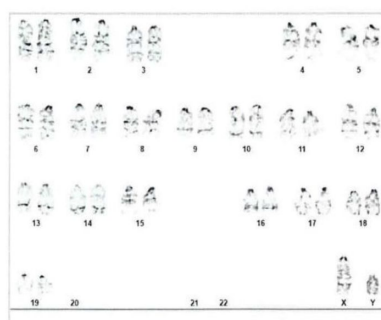


Chromosome Analysis Report: 081055

Date Reported: Wednesday, April 1, 2020
Cell Line: 5-9-1
Passage#: 8
Date of Sample: 3/25/2020
Specimen: Mouse iPSC
Results: 40,XY

Cell Line Sex: Male
Reason for Testing: Study routine characterizat
Investigator: Yu-Huey Lin, UT San Antonio

Nonclonal findings: 40,XY,Del(4C5) 41,XY,+11



Cell: 73
Slide: G03
Slide Type: Karyotype
Total Counted: 20
Total Analyzed: 16
Total Karyogrammed: 15
Band Resolution: >230

Interpretation:

This is a normal karyotype; no clonal abnormalities were detected at the stated band level of resolution.

There are nonclonal findings, listed above, one of which contains a chromosomal aberration (gain of chromosome 11) recurrently acquired in pluripotent stem cell cultures. Nonclonal findings may result from technical artifact, but may be due to a developing clonal abnormality or to low-level mosaicism.

Completed by: Kim Leonhard, CG(ASCP)
Reviewed and Interpreted by: Kaitlin C. Lenhart, Ph.D.

Date: _____ **Sent By:** _____ **Sent To:** _____ **QC Review By:** _____

Limitations: This assay allows for microscopic visualization of numerical and structural chromosome abnormalities. The size of structural abnormality that can be detected is >3-10Mb, dependent upon the G-band resolution obtained from this specimen. For the purposes of this report, band level is defined as the number of G-bands per haploid genome. It is documented here as "band level", i.e., the range of bands determined from the four karyograms in this assay. Detection of heterogeneity of clonal cell populations in this specimen (i.e., mosaicism) is limited by the number of metaphase cells examined, documented here as "% of cells counted".

This assay was conducted solely for listed investigator/institution. The results of this assay are for research use only. Unless otherwise mutually agreed in writing, the services provided to you hereunder by WiCell Research Institute, Inc. ("WiCell") are governed solely by WiCell's Terms and Conditions of Service, found at www.wicell.org/privacyandterms. Any terms you may attach to a purchase order or other document that are inconsistent, add to, or conflict with WiCell's Terms and Conditions of Service are null and void and of no legal force or effect.

Supplementary file 1e

iPSC, EpiLC, and PGCLC Media Components

Stock solutions to prepare before making media.

1. Insulin stock sol. (25 mg/mL), (Sigma I-1882): 100 mg in 4 mL (0.01 M HCl), overnight at 4C aliquot 0.5 mL (total 8 tubes) → -20°C

2. Apotransferrin stock sol. (100 mg/mL), (Sigma T-1147): 500 mg in 5 mL (water), overnight at 4 C aliquot 0.5 mL (10 tubes) → -20°C
3. Progesterone stock sol. (0.6 mg/mL) (Sigma P8783): 10 mg in 1 mL ethanol (10 mg/mL) → 60 µL of stock (10 mg/mL) + 940 µL ethanol → filter through a 0.22 µm filter aliquot 100 µL (10 tubes) → -20°C
4. Putrescine stock sol. (160 mg/mL) (Sigma P5780): 320 mg in 2 mL water → 0.22 filter aliquot 60 µL (~30 tubes) → -20°C
5. Sodium selenite stock sol. (3mM) (Sigma S5261, 10G, MW:172.49) 300 mM: 51.88 mg in 1 mL water → 0.22 filter 10 µL (300 mM stock) + 990 µL water = 3 mM aliquot 50 µL (~20 tubes) → -20°C
6. Activin A stock sol. (50 µg/mL) (120-14, total 10 µg/vial, need more) 20 µg (2 vials) + 400 µL water → 5 µL aliquot (~40 tubes) → -20°C
7. bFGF stock sol. (10 µg/mL) (13256-029, 10 µg/vial) 10 µg (total) in 1 mL (10 mM Tris, pH 7.6 with 0.1 % BSA) → aliquot 20 µL → -20°C
8. Recombinant human BMP4 stock sol. (50 ug/mL) (314-BP-010) 60 ug (total) + 1200 µL (4 mM HCl with 0.1% BSA) → aliquot 105 µL (12 tubes total) → -80°C
9. Stem Cell Factor (SCF) stock sol. (50 ug/µL) (455-MC-010) 10 ug (total) + 200 µL (PBS with 0.1% BSA) → aliquot 21 µL (10 tubes total) → -80°C
10. Recombinant mouse EGF stock sol. (500 ug/mL) (2028-EG-200) 200 ug (total) + 400 µL (PBS with 0.1% BSA) → aliquot 10 µL (60 tubes) → -80°C

iPSC media

N2B27 media for culturing iPSCs in feeder-free conditions

Name	Volume	Catalog Number	Storage Temp
DMEM/F12	3.6 mL	21041-025	(4°C)
Insulin stock sol.	0.5 mL	I-1882	(-20°C)
Apotransferrin stock sol.	0.5 mL	T-1147	(-20°C)
7.5% BSA sol.	330 µL	15260-037	(4°C)
Progesterone stock sol.	16.5 µL	P8783	(-20°C)
Putrescine stock sol.	50 µL	P5780	(-20°C)
Sodium selenite stock sol.	5 µL	S5261	(-20°C)

Supplement table 2.

N2 media components (5 mL)

Name	Volume	Catalog Number	Storage Temp
N2	5 mL	Table 1	(RT)
DMEM/F12	495 mL	21041-025	(4°C)
Neurobasal	480 mL	12348-017	(4°C)
B27 (50X)	10 mL	12587-010	(-20°C)
Pen/Strep	5 mL	15070063	(-20°C)
L-glutamine stock sol.	5 mL	35050061/Glutamax	(4°C)
β-ME stock sol., 50 mM	1.8 mL	21985-023	(4°C)

Make a 40 mL aliquot of N2B27 and store at -80°C.

Supplement table 3.

N2B27 media components (1 L)

Name	Volume	Catalog Number	Storage Temp
N2B27	40 mL	Table 2	(-80°C)
CHIR99021	12 μL	1677-5	(-80°C)
PD0325901	3.2 μL	04-0006-02	(-80°C)
ESGRO	4 μL	ESG1107	(4°C)

Store at 4°C for up to 2 weeks.

Supplement table 4.

N2B27 2i-LIF media components (40 mL)

EpiLC differentiation

Plate coating solution.

1. Human plasma fibronectin (1 mg/mL, store at 4°C) (1 mg/mL, FC010, 1 mg total = 1 mL),

Working solution: For a 12-well plate: dilute 10 μL fibronectin with 600 μL PBS to obtain 16.6 μL/mL fibronectin solution. Pipet the solution into one well of a 12-well plate, incubate for at least 1hr at 37°C and then aspirate the fibronectin solution just before plating iPSCs.

Name	Volume	Catalog Number	Storage Temp
N2B27	5 mL	Table 2	(-80°C)
Activin A stock sol.	2 µL	120-14	(-20°C)
bFGF stock sol.	6 µL	13256-029	(-20°C)
KSR	50 µL	10828-028	(-20°C)

Supplement table 5.

N2 EpiLC differentiation (~5 mL)

PGCLC differentiation

Name	Volume	Catalog Number	Storage Temp
GMEM	8.082 mL	11710-035	(4°C)
KSR	1.5 mL	10828-028	(-20°C)
Non-EAA	100 µL	11140-050	(4°C)
Sodium pyruvate	100 µL	11360070	(4°C)
Pen/Strep	100 µL	15070	(4°C)
L-glutamine	100 µL	35050061/Glutamax	(4°C)
β-ME, 50 mM	18 µL	21985-023	(4°C)

Supplement table 6.

GK15 (10 mL)

Name	Volume	Catalog Number	Storage Temp
GK15	9.878 mL	Table 5	(4°C)
BMP4	100 µL	314-BP-010	(-80°C)
SCF	20 µL	455-MC-010	(-80°C)
LIF	1 µL	ESG1107	(-80°C)
EGF	1 µL	2028-EG-200	(-80°C)

Supplement table 7.

PGCLC differentiation media (10 mL)

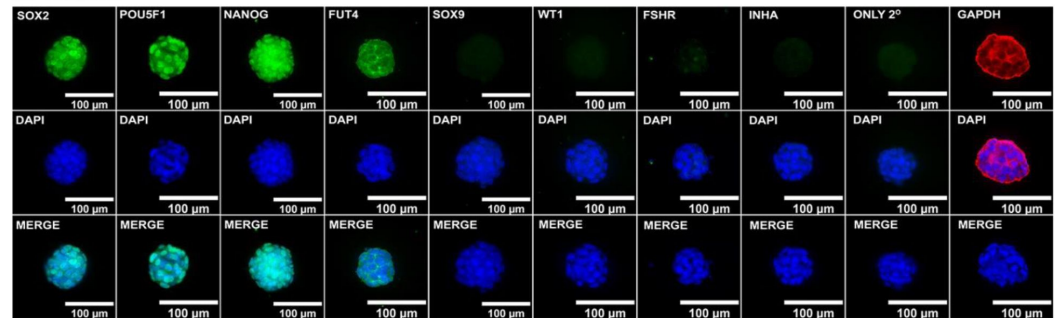
Name	Volume	Catalog Number	Storage Temp
7.5% BSA sol.	1 mL	15260-037	(4°C)
DMEM/F12	74 mL	21041-025	(4°C)

Supplement table 8.

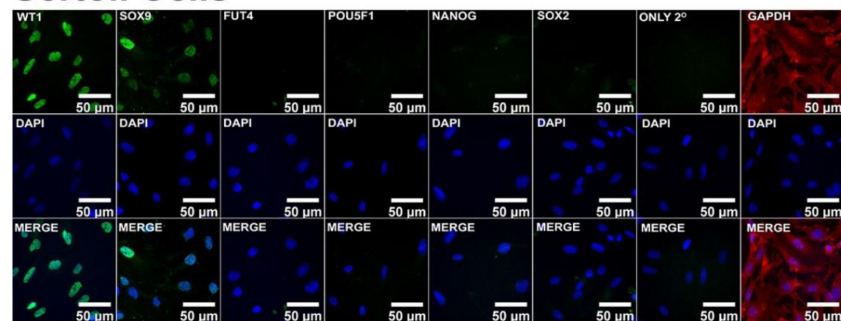
TrypLE wash media (75 mL)

Supplementary file 1f

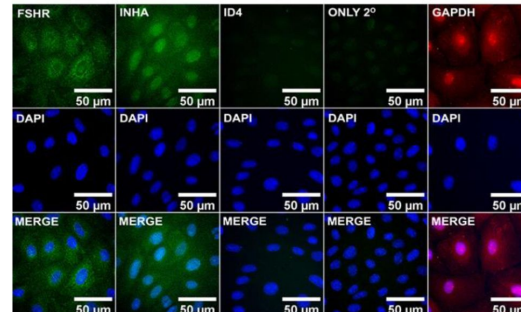
iPSCs



Sertoli Cells



Granulosa Cells



Supplement figure 4.

ICC control staining of cell type-specific markers.

Validation of immunolabeling for known pluripotent and somatic cell type markers in iPSCs, Sertoli Cells, and Granulosa cells along with negative and positive controls.

Supplementary file 1g

Preparation of primary cultures of Sertoli cells from mice protocol

Materials

500 mL DPBS	(Thermo Fisher: Cat. No. 14040133)
DNaseI. 6.64 mg/mL in DPBS	(Sigma: Cat. No. DN-25)
Trypsin 2.5% (10X Gibco)	(Gibco-BRL: Cat. No. 15090-046)
Soybean trypsin inhibitor, 3.4 mg/mL in DPBS	(Gibco-BRL: Cat. No. 17075-011)
Collagenase type IV, 0.70 mg/mL	(Worthington: Cat. No. LS004188)
Sertoli cell media containing retinoic acid	(ScienCell™: Cat. No. 4521)

500 mL DPBS (Thermo Fisher: Cat. No. 14040133) DNaseI. 6.64 mg/mL in DPBS (Sigma: Cat. No. DN-25) Trypsin 2.5% (10X Gibco) (Gibco-BRL: Cat. No. 15090-046) Soybean trypsin inhibitor, 3.4 mg/mL in DPBS (Gibco-BRL: Cat. No. 17075-011) Collagenase type IV, 0.70 mg/mL (Worthington: Cat. No. LS004188) Sertoli cell media containing retinoic acid (ScienCell™: Cat. No. 4521)

Equipment

Pyrex Trypsinizing Flask 300 mL Standard Flask (Fisher Cat. No. 09-552-45)

Procedure: (maintain sterility throughout procedure)

A. Preparation of testes tissue

1. Sacrifice mice by CO₂.
2. Wipe down abdomen with 70% EtOH.
3. Make abdominal incision.
4. Remove the testes and immerse in DPBS in a 50 mL conical tube.
5. Remove tubule bundle from tunica.
 - a. Place testes on parafilm, grasp at one end, and make a small incision in the tunica at the opposite end.
 - b. Push the tubules out of tunica by using the side of closed scissors.
 - c. Weigh flask with fresh DPBS (this is the DPBS volume from the first step below so you must estimate for now (W_1): ____g
 - d. Place tubules in DPBS and weigh (W_2): ____g
 - e. Calculate testes weight (W_T) = ($W_2 - W_1$): ____g.

B. Preparation of Sertoli cells

1. Prepare Erlenmeyer flask by adding:

- Trypsin, $10\times (2.5 \times W_T) = \text{_____ mL}$
- DPBS $(22.5 \times W_T) = \text{_____ mL}$
- DNaseI $(100 \times W_T) = \text{_____ } \mu\text{L}$ (Start with 100 μL of DNaseI and then added 1 mL at the 20 min mark and another 1 mL at the 25 min mark.)
- is very critical, always favor more rather than less and be certain about the concentration when preparing the stock. This along with the flask could be why you are having clumping problems.

Before adding tissue place in a clean petri dish and using a clean razor blade chop downward across the tissue and then rotate the plate 90° and chop across the tissue again until the tissue loses its grape structure as a pre-processing step to help break up the tubules so they digest easily.

Add tissue to the flask. Cover flask with foil, incubate at 37°C for 25-30 min, with occasional swirling. During the last ten minutes, tissue should come apart rapidly. If after 20 min strings seen while swirling (this is very empirical when properly digesting the tissue will break up into pieces approx 2-3 mm in length and be independent of each other not clumped together), add DNaseI at $(50 \times W_T \text{_____ } \mu\text{L})$, after an additional 5 min add DNaseI at $(25 \times W_T \text{_____ } \mu\text{L})$.

- When tubules are separated, move flask to sterile hood, let tubules settle for 5 mins. Then remove and discard overlying liquid. The solution will be very sticky and the tissue will want to come out with the supernatant. Don't lose too many cells here, if you can not get all the supernatant off just go ahead and add the trypsin inhibitor, but use another 0.5 mL on top of what you have figured to compensate for the increased volume.
- Add trypsin inhibitor $(2.5 \times W_T) = \text{_____ mL}$. Swirl and incubate for 1 min.
- Add $(25 \times W_T) = \text{_____ mL}$ fresh DPBS, swirl, allow to settle. (Wait to add this if you did not get much of the supernatant off before, too much volume will increase the buoyancy and not let the cells settle out.
- Remove supernatant, add $(20 \times W_T) = \text{_____ mL}$ DPBS, swirl, allow to settle.
- Remove supernatant, add
 - $(7.5 \times W_T) = \text{_____ mL}$ DPBS
 - Collagenase $(2.5 \times W_T) = \text{_____ mL}$
 - DNaseI $(25 \times W_T) = \text{_____ } \mu\text{L}$ (Add in 1mL here of DNaseI instead if below 1 mL)
- Incubate in water bath at 37°C for 10 min. After 5 min, swirl continuously. (During the last 3 min aliquots are checked by microscope to monitor the release of peritubular cells. A drop of suspension is placed on a glass slide and the tissue is viewed under a coverslip at 10x mag. The coverslip is moved slightly and as the tissue moves peritubular cells can be seen to peel off. The tissue is very fragile at this stage and must not be allowed to over digest. Digestion is complete when the surface of large fragments no longer appears smooth.
- Transfer tubules to 50 mL conical tube, wash the flask with DPBS, and centrifuge at 200xg for 2 min.
- Remove supernatant, add 30-35 mL DPBS, and centrifuge again.
- Repeat step 9.
- Remove supernatant and resuspend and plate cells in Sertoli cell media containing retinoic acid from ScienCellTM.
- Cells are incubated at 37°C in 5% CO_2 , in a humidified chamber. Medium is changed every 48 hours after 24-36 hours shock the cells with 10 mM Tris-HCl, pH 7.4 for 2 min. This will remove most of the contaminating germ cells. Make this media by dissolving 12.1g of Trisbase in 80 mL of ddH_2O and adjust the pH to 7.4 with HCl. This will make a 1M stock. To make a 10 mM solution, add 1mL of this 1M stock to 99 mL of ddH_2O and filter sterilize.

Supplementary file 1h

Preparation of primary cultures of granulosa cells from mice protocol

Materials

70% EtOH	(Fisher: Cat. No. BP8203)
500 mL DPBS	(Thermo Fisher: Cat. No. 14040133)
BSA	(Sigma: Cat. No. A9085)
Collagenase type IV	(Worthington: Cat. No. LS004188)
Glass Pipettes	(Fisher: Cat. No. 13-678-20A)
DMEM/F12	(Gibco Cat. No. 21041-025)
FBS	(Gibco Cat. No. 10082-147)
Penicillin/Streptomycin 100X	(Gibco Cat. No. 15140-122)
0.25% Trypsin-EDTA	(Gibco Cat. No. 25200-072)

Procedure

A. Preparation of media

1. Prepare 50 mL of DPBS with 0.5% BSA by dissolving 0.25 g of lyophilized BSA powder into 50 ml of DPBS medium.
2. Collagenase type IV solution (540 U/pair of ovaries) in 3 mL of DPBS.
3. Prepare a glass pipette with a pointed tip that you can use to transfer the follicles. Set up an alcohol flame. Holding a glass pipette from both ends place the thicker section of glass just before the pipette tapers off over the hottest part of the flame. Rotate the tubing to make sure it is evenly heated. Continue to heat the glass until it starts to melt and becomes flexible. Then, with a quick motion, pull the glass so it forms a thin tube section. This will take some trial-and-error force that will give you the size that works best. Etch the glass and tap near the etched section to make a clean break in the tube. Finally, sterilize the tube and remove jagged edges by quickly running the completed tube through the flame. Attach the glass pipette to some form of controlled suctioning device (i.e. tubing connected to small syringe)
4. Prepare 50 mL of DMEM/F12 with 15% FBS by adding 7.5 mL of FBS and 0.25 mL of 100X Penicillin/Streptomycin stock solution to 42.25 mL of DMEM/F12 medium.

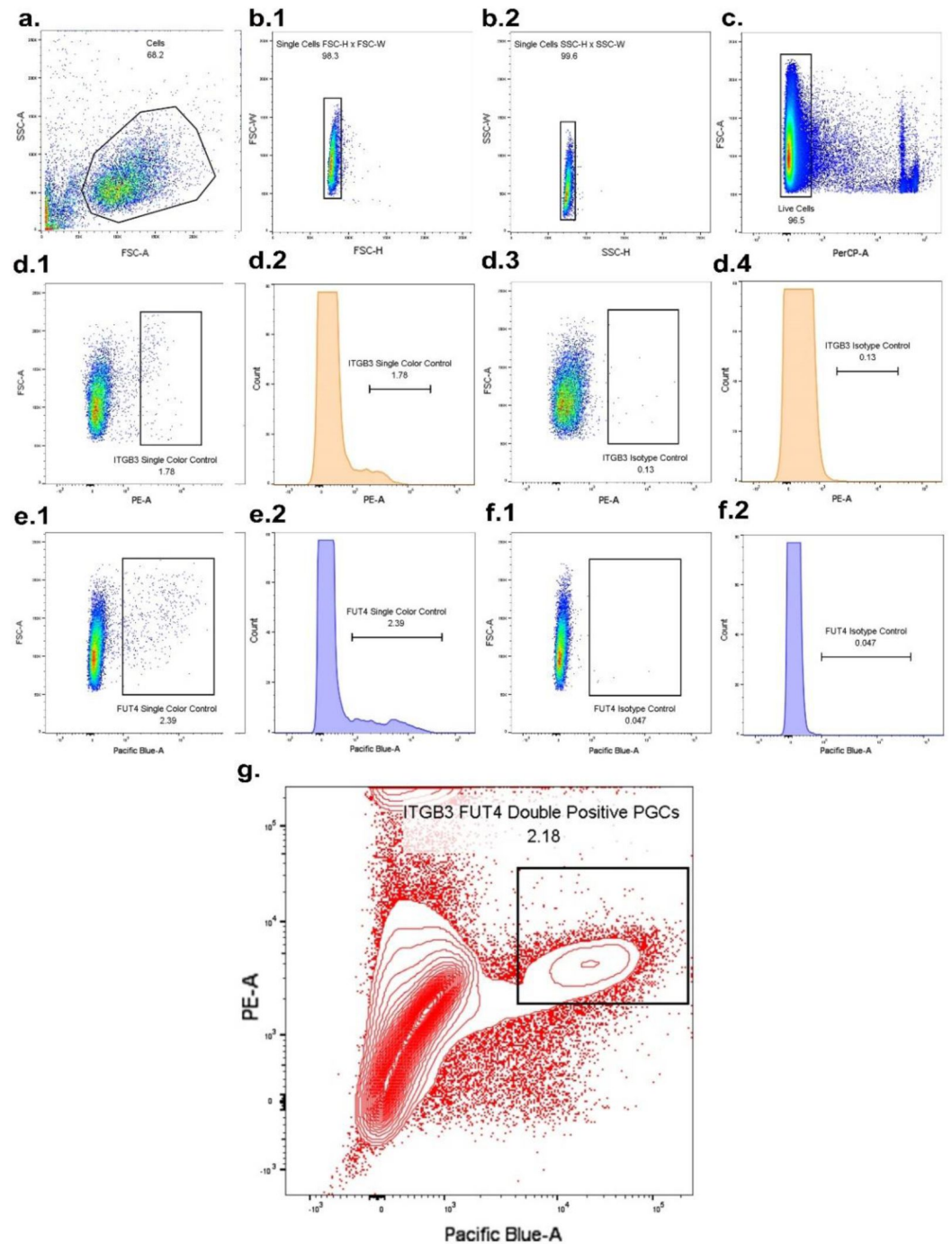
B. Isolation of ovarian follicles from mice

1. Sacrifice mice by CO₂.
2. Wipe down abdomen with 70% EtOH.
3. Make abdominal incision.
4. Remove the ovaries and immerse in DPBS in petri dish.
5. Decap the ovaries by resecting away fat and connective membranes that surrounds and protect the ovary.
6. Transfer the decapped ovaries to a clean petri dish with DPBS and no BSA. This will help to clean off any contaminating hairs, or fat and blood droplets released while decapping the ovaries.
7. Section the ovaries into fourths by cutting them with the needle and forceps. This will help the collagenase digest the tissue and release follicles in the next step.
8. After the ovaries have been divided into fragments add them to the activated collagenase digest solution and let the ovaries digest for 25 mins under constant agitation.
9. At the end of 25 mins add in DPBS containing 0.5% BSA to inactivate the collagenase then centrifuge the tube at 60xg for 5 mins to pellet the follicles²¹⁴.
10. After the tissue digest has finished spinning remove the supernatant and resuspend in 7 mL of DPBS with 0.5% BSA and transfer to a new petri dish to individually pick out follicles.

C. Preparation of granulosa cells

1. Take the petri dish containing resuspended ovarian tissue and place it under stereomicroscope.
2. Move the preantral follicles over to the clean droplet of DPBS with 0.5% BSA. It's best to move only a few follicles over at a time as pulling up many follicles can cause them to sediment and stick to the inside of the glass pipette if they are pulled up too far inside.
3. Once you have moved over enough follicles, set up a 1.5 mL microcentrifuge tube containing 1 mL of 0.25% activated trypsin.
4. Gently swirl the dish containing the preantral follicles to get them to clump together in the center and collect them with a 1000 µL micropipette before transferring them to the droplet containing the trypsin EDTA.
5. Let the follicles digest for 5 minutes. Pipette the digesting cells to help break up the follicles during the incubation period.
6. Spin tube at 1000xg for 5 mins to pellet the granulosa and oocyte cells.
7. Remove the supernatant and resuspend pellet in DMEM/F12 supplemented with 15% FBS and plate cells.

Supplementary file 1i

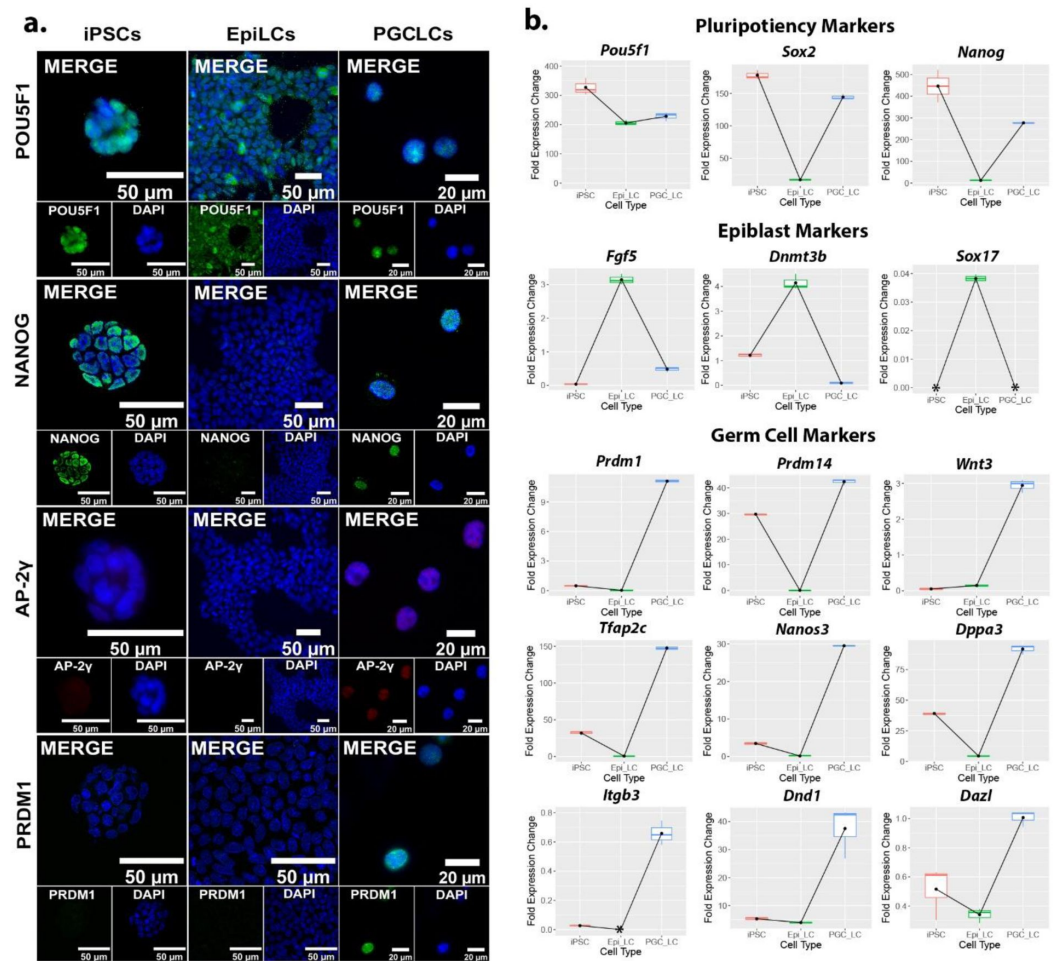


Supplement figure 5.

FACS sorting for ITGB3/FUT4 enriched primordial germ cell like cells.

a) Gating for cells. **b.1-2)** Gating for singlet cells. **c)** Gating for live cells. **d.1-2)** Single color control ITGB3 positive cells. **d.3-4)** IgG isotype control. **e.1-2)** Single color control FUT4 positive cells. **f.3-4)** IgG isotype control. **g)** Sorting for PGCLC-enriched ITGB3/FUT4 double positive population (2.18% of total cells)

Supplementary file 1j



Supplement figure 6.

Relative expression of markers for PGCLC induction from iPSCs.

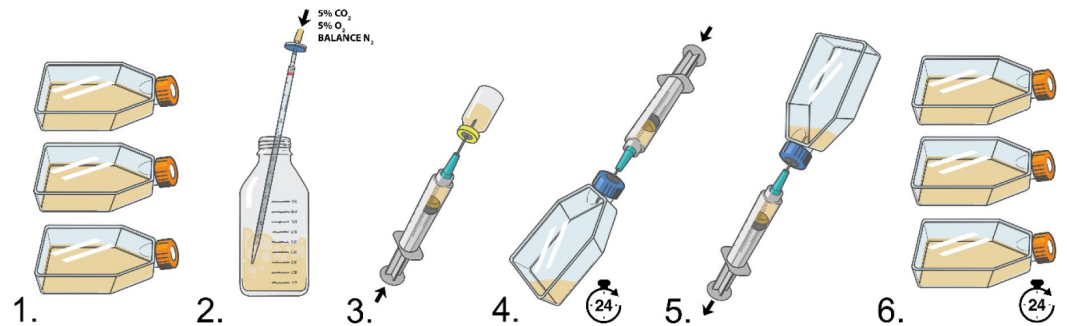
a) ICC of pluripotency and germ cell marker expression throughout the transition from iPSCs to PGCLCs. **b)** qRT-PCR of pluripotency, epiblast, & germ cell markers indicating gene expression profiles during induction of PGCLCs from iPSCs. Each gene fold expression is relative to the housekeeping gene *Gusb* using the $\Delta\Delta C_q$ method. The symbol * indicates that the expression of transcripts in the sample were either non-existent or so low as to be undetectable by qRT-PCR.

Gene	Forward	Reverse	Efficiency
<i>Pou5f1</i>	TTCAGCCAGACCACCATCTGT	GGTTCTCATTGTTGTCGGCTTC	106.02%
<i>Nanog</i>	GTGCTTGCTTGCCTTGGAG	AGCTGGCATCGGTTTCATCAT	103.86%
<i>Tfap2c</i>	AGGCATCTCATTCTGCGGG	CACGGACAGGCTTAGAGGTC	101.37%
<i>Prdm1</i>	GAAAAATGGGAGCCCCGACA	AAGACGGAAGGGGACTGTGA	92.60%
<i>Prdm14</i>	ACAGCCAAGCAATTTGCACTAC	TTACCTGGCATTTTCATTGCTC	104.57%
<i>Sox2</i>	TTGGGAGGGGTGCAAAAAGA	TTCTAGTCGGCATCACGGTT	98.20%
<i>Sox17</i>	CAAAGCGAAGATTGCAGGGT	TTCCAGTCCCTGGTCAGTC	107.78%
<i>Fgf5</i>	TACGTGGCCCTGAACAAGAG	TGGAACAGTGACGGTGAAGG	108.72%
<i>Dnmt3b</i>	GCTTTGCTTCACTGGGTCTC	AACTGATGGGGTACTGACGC	98.12%
<i>Wnt3</i>	CCTGGACCACATGCACCTAAA	CGGAGGCACTGTCGTACTTG	95.11%
<i>Nanos3</i>	CACTACGGCCTAGGAGCTTGG	TGATCGCTGACAAGACTGTGG	90.56%
<i>Dppa3</i>	TGAACGGGACAGTGAGCCA	AGCCAGGGCAGCGTACAAT	99.12%
<i>Itgb3</i>	ACGGGACTTTTGAGTGTTGG	GCTGAGAGGGTCGGTAATCC	100.54%
<i>Dnd1</i>	AATGGGTAAAGCAGAGCCTGT	AGGGCAAGGTTCTCACAAC	96.18%
<i>Dazl</i>	GCAGAGAACCTTTGTACCCAC	TAGCAAGGTGCTGTTCCAGTG	102.75%
<i>Gusb</i>	GCAGCCCTTCGGGACTTTAT	TCCTCAACACCACTCTCATGTC	97.17%

Supplement table 9.

Primer designs

Supplementary file 1k

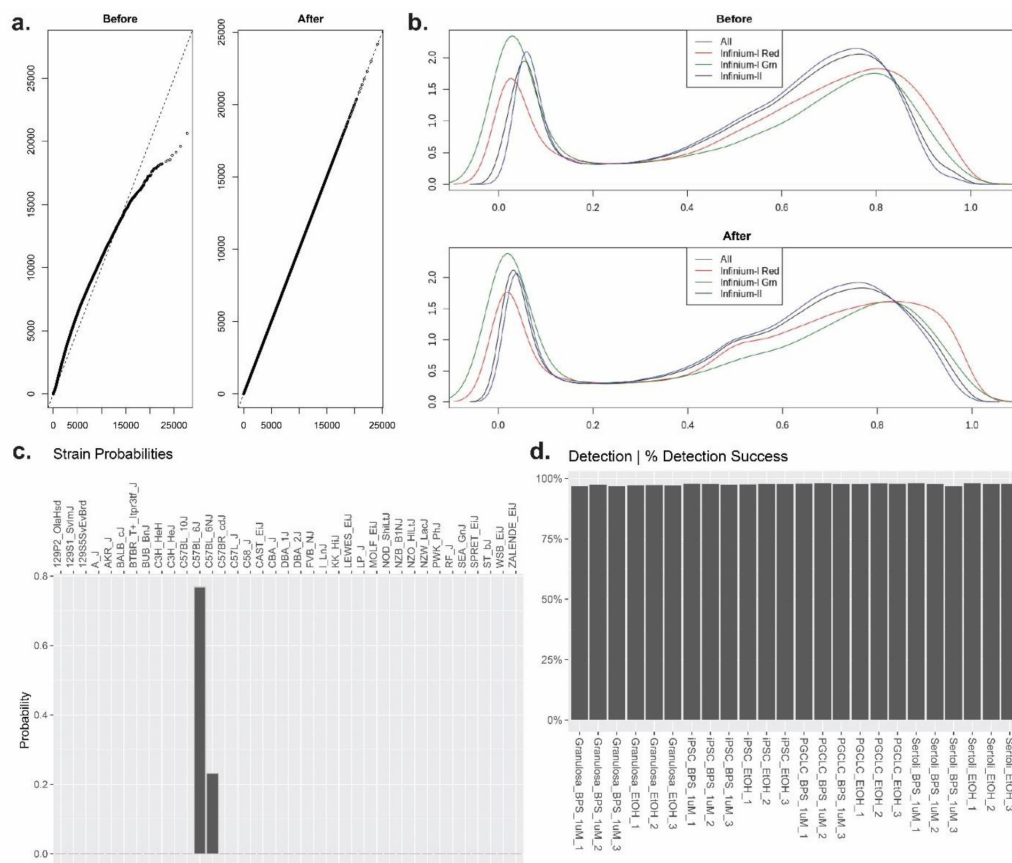


Supplement figure 7.

Chemical exposure experimental workflow. 1)

Cells are passaged into T-25 flasks with filter caps. **2)** Mixed blood gas (carbon dioxide 5%, oxygen 5%, and balance nitrogen) is filtered and bubbled into media to prepare media to be mixed with diluted chemical treatment and added to air-tight cell culture flasks. **3)** Media is transferred into glass vials and diluted chemical is added. **4)** T-25 filter caps are replaced with air-tight caps with septums and cell media containing chemicals is added via syringe and left for 24 hours. **5)** After 24 hours, chemical-containing media is removed and cells are washed with buffer. **6)** T-25 air-tight septum caps are replaced with filter caps and cells are cultured for an additional 24-hour "chase" period.

Supplementary file 1m

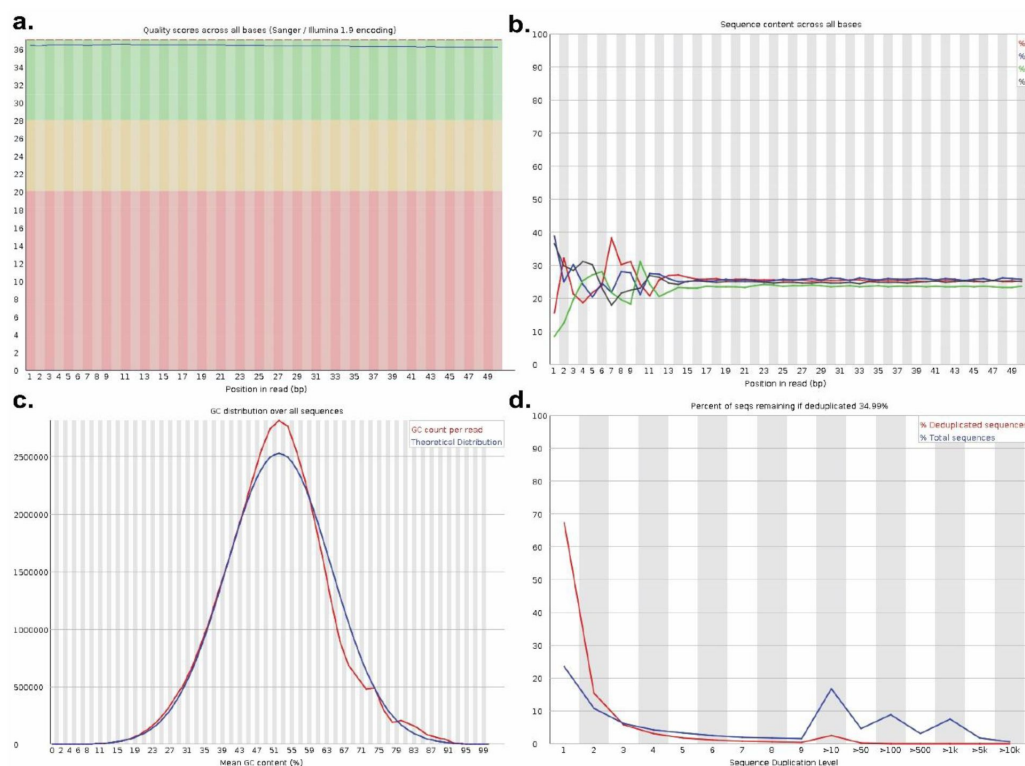


Supplement figure 8.

Quality control for Infinium Mouse Methylation BeadChip Array data.

a) Non-linear correction of dye bias removal from sample data. **b)** Background signal subtraction from samples to limit noise. **c)** Prediction of correct C57B6 mouse strain from samples included in the study based on built-in controls on Infinium Mouse Methylation BeadChip Array. **d)** Average CpG probe detection success of 97.58% percent across all samples indicating efficient bisulfite conversion of all samples.

Supplementary file 1n



Supplement figure 9.

Quality control for RNA-seq data.

RNA-seq quality control data from a one of the three PGCLC replicates exposed to BPS as an exemplar **a)** Base calls had high quality scores (phred scores > 30) for all bases in reads. **b)** Reads showed equal distributions of all four bases following the initial adaptor sequence and sufficient base complexity. **c)** Distribution of GC sequences across reads aligns very closely with the theoretical distribution. **d)** Duplication plot indicates deduplicated libraries contained ~67% unique sequences which indicates sufficient library complexity for subsequent downstream data processing.

Supplementary file 1o

Sample	Insert Size (bp)	Median Coverage	Read Counts (Millions)	Aligned Reads (%)	Duplicate Reads (%)	GC (%)
iPSC_E_1	320	37 X	851.9	99.0	28.6	22
iPSC_E_2	374	33 X	752.6	99.1	26.4	22
iPSC_E_3	378	38 X	873.5	99.2	27.2	22
iPSC_BPS_1	360	28 X	626.7	99.1	23.6	22
iPSC_BPS_2	330	30 X	695.3	99.1	26.6	22
iPSC_BPS_3	338	33 X	755.8	99.1	27.7	22
PGCLC_E_1	388	32 X	707.9	99.0	25.6	23
PGCLC_E_2	374	25 X	559.4	99.0	24.0	22
PGCLC_E_3	412	30 X	673.5	99.2	24.9	22
PGCLC_BPS_1	399	31 X	672.9	99.1	25.6	22
PGCLC_BPS_2	371	32 X	753.3	99.1	28.6	22
PGCLC_BPS_3	392	27 X	599.3	99.1	23.6	22

Supplement table 10.

Quality control metrics for EM-seq data.

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Editors

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Reviewer #1 (Public Review):**Summary:**

The goal of this study was to use in vitro cell populations to determine mechanisms that may be important for the propagation of epimutations induced by EDCs in vivo. To do this, authors exposed induced pluripotent stem cells (iPS), somatic cells (Sertoli, granulosa), and primordial germ cell like cells (PGCLCs) to BPS, and conducted epigenomic and transcriptomic analyses on outcomes. The importance of estrogen receptors, and the relationship of epigenomic results to genomic sites expressing EREs, were also determined in the different cell types. Results revealed differential effects of BPS in each cell population on each of these endpoints, and that epimutations were prevalent in enhancer regions with EREs with the exception of PGCLCs (which do not express ERs). The authors speculate that because epimutations also occurred in regions without EREs, especially in PGCLCs, other mechanisms may be in place. Finally, epimutations induced in iPSCs exposed to BPS that were subsequently differentiated into PGCLCs demonstrated that most epimutations were corrected.

Strengths:

A strength of this work is the use of different cell types representing somatic cells that would be the major recipient of EDC exposure; pluripotent cells representing preimplantation embryos; and PGCLCs that model the early germline in which epigenetic reprogramming takes place. Work differentiating the iPSCs from PGCLCs with or without BPS exposure at the iPSC level is also very informative as it suggests that most epimutations are corrected, at least in vitro. The paper is well-written and studies were technically well-executed and validated. Results are novel and likely to be of interest to those interested in transgenerational inheritance of environmentally-induced traits, as well as others more broadly interested in epigenetic mechanisms.

Critique/Weaknesses:

- (1) A problem with in vitro work is that homogeneous cell lines/cultures are, by nature, absent from the rest of the microenvironment. The authors need to discuss this.
- (2) What are n's/replicates for each study? Were the same or different samples used to generate the data for RNA sequencing, methylation beadchip analysis, and EM-seq? This clarification is important because if the same cultures were used, this would allow comparisons and correlations within samples.
- (3) In Figure 1, it is interesting that the 50 uM BPS dose mainly resulted in hypermethylation whereas 100 uM appears to be mainly hypomethylation. (This is based on the subjective appearance of graphs). The authors should discuss and/or present these data more quantitatively. For example, what percentage of changes were hypo/hypermethylation for each treatment? How many DMRs did each dose induce? For the RNA-seq results, again, what were the number of up/down-regulated genes for each dose?
- (4) Also in Figure 1, were there DMRs or genes in common across the doses? How did DMRs relate to gene expression results? This would be informative in verifying or refuting expectations that greater methylation is often associated with decreased gene expression.
- (5) In Figure 2, was there an overlap in the hypo- and/or hyper-methylated DMCs? Please also add more description of the data in 2b to the legend including what the dot sizes/colors mean, etc. Some readers (including me) may not be familiar with this type of data presentation. Some of this comes up in Figure 4, so perhaps allude to this earlier on, or show these data earlier.

(6) iPSCs were derived from male mice MEFs, and subsequently used to differentiate into PGCLCs. The only cell type from an XX female is the granulosa cells. This might be important, and should be mentioned and its potential significance discussed (briefly).

(7) EREs are only one type of hormone response element. The authors make the point that other mechanisms of BPS action are independent of canonical endocrine signaling. Would authors please briefly speculate on the possibility that other endocrine pathways including those utilizing AREs or other HREs may play a role? In other words, it may not be endocrine signaling independent. The statement that the differences between PGCLCs and other cells are largely due to the absence of ERs is overly simplistic.

(8) Interpretation of data from the GO analysis is similarly overly simplistic. The pathways identified and discussed (e.g. PI3K/AKT and ubiquitin-like protease pathways are involved in numerous functions, both endocrine and non-endocrine. Also, are the data shown in Figure 6a from all 4 cell types? I am confused by the heatmap in 6c, which genes were significantly affected by treatment in which cell types?

(9) In Figure 7, what were the 138 genes? Any commonalities among them?

(10) The Introduction is very long. The last paragraph, beginning line 105, is a long summary of results and interpretations that better fit in a Discussion section.

(11) Provide some details on husbandry: e.g. were they bred on-site? What food was given, and how was water treated? These questions are to get at efforts to minimize exposure to other chemicals.

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Reviewer #2 (Public Review):

Summary:

This manuscript uses cell lines representative of germ line cells, somatic cells, and pluripotent cells to address the question of how the endocrine-disrupting compound BPS affects these various cells with respect to gene expression and DNA methylation. They find a relationship between the presence of estrogen receptor gene expression and the number of DNA methylation and gene expression changes. Notably, PGCLCs do not express estrogen receptors and although they do have fewer changes, changes are nevertheless detected, suggesting a noncanonical pathway for BPS-induced perturbations. Additionally, there was a significant increase in the occurrence of BPS-induced epimutations near EREs in somatic and pluripotent cell types compared to germ cells. Epimutations in the somatic and pluripotent cell types were predominantly in enhancer regions whereas that in the germ cell type was predominantly in gene promoters.

Strengths:

The strengths of the paper include the use of various cell types to address the sensitivity of the lineages to BPS as well as the observed relationship between the presence of estrogen receptors and changes in gene expression and DNA methylation.

Weaknesses:

The weaknesses include the lack of reporting of replicates, superficial bioinformatic analysis, and the fact that exposures are more complicated in a whole organism than in an isolated cell line.

